

The birth–death–catastrophe process

Distribution theory, quasi-stationarity, and burst statistics

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Chapter 5

The birth–death–catastrophe process: distribution theory, quasi-stationarity, and burst statistics

5.1 Introduction

The standard models of within-host infection treat the infected cell as a black box. In the Basic Model of Viral Replication, a cell that has been infected releases new particles at a constant rate p until it is removed at a constant rate d_I ; nothing about the cell’s interior enters the dynamics, and the two constants are phenomenological — parameters to be fitted, not quantities to be derived [1, 2, 3]. For a budding virus this is at least a defensible caricature. For a lytic pathogen it is plainly wrong in mechanism, whatever it may achieve by fitting: such a pathogen grows inside the cell, and releases nothing until the cell ruptures, at which point it releases everything at once. *Y. pestis* in the macrophage, *F. tularensis* in the macrophage, the lytic phage in its bacterium — all follow the intracellular replication-and-lysis picture, and all are poorly served by a constant drip.

Any replacement of that black box must begin with a complete description of a single infected cell, and that description is the subject of this chapter. The intracellular model was set up in the chapter on the birth–death–catastrophe process: the birth–death–catastrophe process, in which each pathogen particle inside one cell divides at rate λ , dies at rate μ , and triggers the rupture of the whole cell at rate δ . That chapter derived the principal single-cell quantities — the no-rupture probability $I(t)$, the non-fixation probability $I_{\text{fix}}(t)$, the mean and second moment $J(t)$, $K(t)$, and the cumulative mean release $V(t)$ — each in closed form. What it did not derive is the full distribution of the process, or the statistics of the rupture event itself.

One question organises what follows. *What is the complete law of a single bursting cell?* The answer is a single geometric family whose ratio slides towards a fixed endpoint, and the burst inherits that endpoint. That inheritance is the chapter’s central identity, and it is not obvious: one law comes from conditioning a surviving trajectory on never being absorbed, the other from averaging over the instant each trajectory ruptures, and the two constructions share no step. [Section 5.6](#) resolves it. Two probes then ask how far the answer reaches: several founders sharing one cell, and one cell’s burst founding the next. Both are stress tests rather than extensions, and both mark a boundary.

The derivation proceeds as follows.

- The load is geometric at every time, with a ratio $P(t)$ that slides from 0 to $1/a$.
- The quasi-stationary law is that endpoint.
- The burst, averaged across rupture times, is the same law.
- Two probes — several founders sharing one cell, then one cell's burst founding the next — test how far the structure reaches.

The chapter is organised around that derivation. [Section 5.2](#) recaps the chapter on the birth–death–catastrophe process. [Sections 5.3](#) and [5.4](#) produce the generating function and the geometric family. [Sections 5.5](#) and [5.6](#) take the endpoint and the identity. [Sections 5.7](#) and [5.8](#) give the times and the three means. [Sections 5.9](#) and [5.10](#) run the two probes. [Section 5.11](#) returns to the constant drip, at the level where the objection was made: one cell.

Two conventions for the rupture transition are current. The ruptured process may be sent to state 0, which we call *killing*, or to a separate absorbing state H , which we call *catastrophe*. This chapter works in the catastrophe convention throughout, and nothing is lost by doing so: the two generating functions differ by a function of time alone, exactly and for every z , as [proposition 5.3](#) records.

5.1.1 Standing assumptions

Five assumptions are load-bearing, and all five are contestable.

1. **Linear birth.** Each particle divides at rate λ whatever the current load: there is no intracellular carrying capacity. Without it the Riccati equation [\(5.7\)](#) is not autonomous and no closed form survives.
2. **Catastrophe hazard exactly proportional to load.** Rupture is triggered at rate δn from load n — not δn^2 , not a threshold at a critical load, not a maturation-timed lysis. This is the strongest assumption in the chapter and it is what makes everything solvable: the burst-size law of [theorem 5.9](#) is an integral against the hazard δk , and its telescoping depends on that hazard being linear.
3. **Instantaneous, complete release.** The whole load leaves the cell at one instant, which is what makes the burst size a single random variable $\mathcal{K}$ rather than a release process.
4. **Constant rates.** No eclipse phase and no age-dependence within the cell, so that the process is time-homogeneous and the generating function depends on t only through $w = e^{\theta t}$.
5. **Independent lineages.** Descendants of distinct particles evolve independently until the shared catastrophe clock rings. This is used decisively at $G_k = G^k$ in [\(5.23\)](#) and throughout [section 5.9](#).

The first is revisited as an open problem in [section 5.12.4](#); the second is what [remark 5.11](#) isolates as the criterion behind the chapter's central identity.

5.1.2 Contributions

Not all of this chapter is background. The mathematical introduction defines quasi-stationarity and the Yaglom limit, takes existence from the literature, and computes the limiting conditional *mean*, noting explicitly that convergence of moments does not establish convergence of the full conditional law. It leaves open whether a quasi-stationary distribution survives when the process is conditioned on *neither* extinction nor rupture. Five results are new to this chapter.

1. The full conditional law exists in closed form for all $\mu \geq 0$ and is geometric with ratio $1/a$ (theorems 5.4 and 5.6).
2. Its mean was already known to equal the mean burst size. The two *laws* are identical, and the identity has a structural proof: size-biasing telescopes the occupation integral onto the endpoint of the process's own geometric ratio (theorem 5.10).
3. The chapter on the birth–death–catastrophe process gives the marginal law of the burst time. Conditioned on the burst size, that law is a harmonic-number law with a Gumbel limit (proposition 5.12).
4. The single-founder release kernel was already in hand. For k founders sharing one cell the kernel g_k is superlinear in k , while the lifetime yield is sub-extensive, (5.72) and (5.74).
5. Chained immediate transfer has an exact negative-binomial structure, (5.79).

Everything else is standard, or is quoted from the chapter on the birth–death–catastrophe process, and section 5.A marks which is which. One argument in the chapter is heuristic rather than complete — the interchange of summation and integration in the hypergeometric machinery of section 5.C.2 — and it is marked where it occurs.

5.1.3 What this chapter exports

The chapter is a supplier as well as an argument. The chapter on population dynamics needs five things from here: $I_{\text{fix}}(a)$ and $g(a) = \delta K(a)$ as the survival and release kernels of a renewal model; g_k as the kernel for a cell founded at multiplicity k ; the mean productive lifetime $\mathbb{E}[T_{\text{prod}}]$ together with the matched constant-rate death rate $d_{\mathcal{T}} = 0.351$ it implies; and $\mathbb{E}[W_t^2]$, for the between-well variance that a limiting-dilution assay measures. Each is derived below and flagged where it is derived.

One parameter set recurs. Unless stated otherwise, illustrations use $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, for which $a = 1.0616$, so $b = 0.1884$ and $\mathbb{E}[K \mid \text{burst}] = 17.23$: most cells burst, and the bursts are large. Those rates sit on the *F. tularensis* side of the model's range, where $a = 1.001$, rather than the *B. anthracis* side, where $a = 2.66$, with the numbers compressed for legibility. The figures may therefore be read as a rescaled lytic cell rather than an arbitrary choice.

5.2 The process, its roots, and its closed forms

The birth–death–catastrophe (BDC) process was defined and solved in the chapter on the birth–death–catastrophe process. Four things are needed from it here: the notation, the process and its release counter, the roots, and the identity relating them. Section 5.A collects the closed forms.

5.2.1 Canonical notation

Table 5.1 fixes the notation, taken unchanged from the chapter on the birth–death–catastrophe process. Every symbol appearing in more than one section is listed; those local to a single derivation are introduced where they occur.

5.2.2 The process and its release counter

The object throughout is one cell holding one intracellular population.

Symbol	Meaning	Notes
λ, μ, δ	birth, death, catastrophe rates	per capita
X_t	intracellular load	a CTMC on $\{0, 1, 2, \dots\} \cup \{H\}$
H	absorbing rupture state	
W_t	released count from one cell	0 until the burst
τ	burst time	possibly ∞
K	burst size	never bare K , since $K(t) = \mathbb{E}[X_t^2]$
$G(z, t)$	probability generating function (PGF) of the load	defective under catastrophe
G_k	PGF from k founders	$= G^k$
$p_n(t)$	$\mathbb{P}\{X_t = n\}$	
$P(t), \sigma$	geometric ratio of the load; the clock $\sigma = P(t)$	slides from 0 to $1/a$
ξ, s	characteristic coordinate and parameter	$t = -\xi$; section 5.3 . In section 5.10 , $s = \delta/(\lambda + \delta)$
$I(t), D(t)$	no burst by t ; cleared internally, no burst	$D = p_0(t)$
$I_{\text{fix}}(t)$	$\mathbb{P}\{X_t \notin \{0, H\}\} = I - D$	productive infection
$J(t), K(t)$	$\mathbb{E}[X_t], \mathbb{E}[X_t^2]$	
$V(t)$	$\mathbb{E}[W_t]$	cumulative mean release
$\varphi(t)$	density of τ	defective, mass $1 - b$
a, b	characteristic roots	$b < 1 < a$
A, B, θ	$a - 1, 1 - b, \lambda(a - b)$	$AB = \delta/\lambda$
κ, w	$1 + \delta/(2\lambda); e^{\theta t}$	in $K(t)$; the recurring exponential
q_n, ν	quasi-stationary law	ν when read as a vector
$\vartheta, \mathcal{Q}$	decay rate; killed subgenerator	$\vartheta = \theta$; proposition 5.7
T_{prod}, H_n	productive lifetime; harmonic number $\sum_{j \leq n} 1/j$	
$g_k(t)$	release flux from k founders	$= \delta K_k(t)$
r_k, t_k, T_k	size, time and interval of the k th rupture	section 5.10
ρ	$\lambda/(\lambda + \delta) = 1 - s$	section 5.10
$d_{\mathcal{I}}, p$	budding death and release rates	section 5.11
${}_2F_1$	Gauss hypergeometric function	

Table 5.1: Canonical notation for this chapter.

Definition 5.1 (Birth–death–catastrophe process). The load X_t is a continuous-time Markov chain on $\{0, 1, 2, \dots\} \cup \{H\}$ which, from any state $n \geq 1$, jumps

$$\begin{aligned}
n &\rightarrow n + 1 && \text{at rate } \lambda n \quad (\text{birth}), \\
n &\rightarrow n - 1 && \text{at rate } \mu n \quad (\text{death}), \\
n &\rightarrow H && \text{at rate } \delta n \quad (\text{catastrophe}).
\end{aligned}$$

Both 0 and H are absorbing: the population either clears itself internally, or the cell ruptures. Unless stated otherwise $X_0 = 1$.

The classical treatments [\[4, 5, 6\]](#) allow immigration and distributed catastrophe sizes; this specialisation is the one with the closed-form family we need. Rupture is what matters, so the release count runs alongside the load: with τ the catastrophe time,

$$W_t = \begin{cases} 0, & t < \tau, \\ X_{\tau-}, & t \geq \tau, \end{cases} \quad (5.1)$$

so that at rupture the whole pre-burst load is dumped and then held. [Figure 5.1](#) shows one path of the pair; every single-cell release quantity below is a functional of (X_t, W_t) .

5.2.3 Roots and the workhorse identity

Write

$$\eta = \frac{\lambda + \mu + \delta}{2\lambda}, \quad a = \eta + \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad b = \eta - \sqrt{\eta^2 - \frac{\mu}{\lambda}}, \quad (5.2)$$

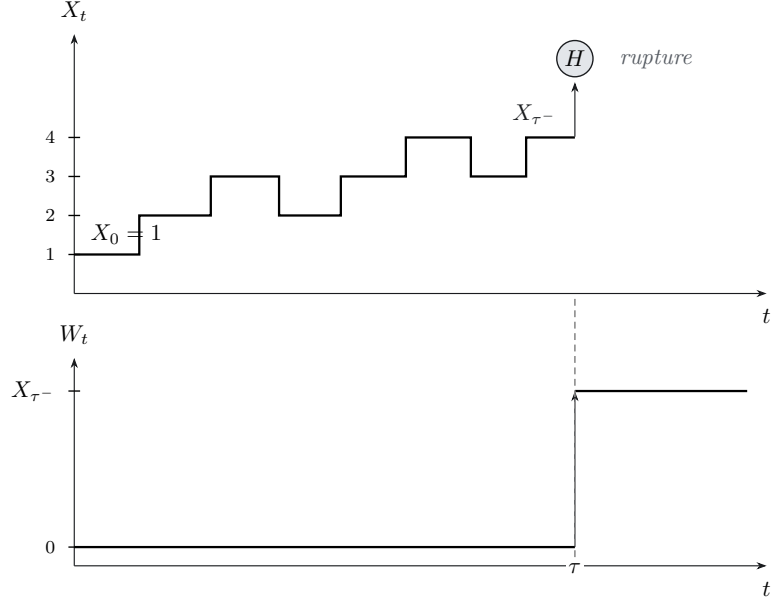


Figure 5.1: A representative path of the joint process (X_t, W_t) . The intracellular load follows a birth–death path until rupture at time τ , when X_t enters the absorbing state H . At the same instant the released count jumps from zero to the pre-rupture load $X_{\tau-}$ and remains there. The two axes share a time origin: nothing leaves the cell before τ , and everything leaves at τ .

the two roots of $\lambda f^2 - (\lambda + \mu + \delta)f + \mu$, and set

$$A := a - 1 > 0, \quad B := 1 - b > 0, \quad \theta := \lambda(a - b) = \sqrt{(\lambda + \mu + \delta)^2 - 4\lambda\mu}. \quad (5.3)$$

Vieta's relations give $ab = \mu/\lambda$ and $a + b = (\lambda + \mu + \delta)/\lambda$, and the identity doing most of the algebraic work below is

$$AB = (a - 1)(1 - b) = (a + b) - ab - 1 = \frac{\delta}{\lambda}, \quad A + B = a - b. \quad (5.4)$$

That $b < 1 < a$ follows immediately from $\lambda(1 - a)(1 - b) = \lambda(1 - (a + b) + ab) = -\delta < 0$, which places 1 strictly between the roots.

The two-type chapter solves a Riccati equation of exactly this shape under unrelated names: roots that straddle one, a root gap playing the part of $a - b$, and the same factorisation. [Section 5.12.5](#) records the correspondence rather than acting on it.

5.2.4 The closed forms

There are three absorbing-behaviour probabilities, not two.

Definition 5.2 (Fixation functions).

$$I(t) = \mathbb{P}\{\text{no burst by } t\}, \quad D(t) = \mathbb{P}\{\text{internal extinction by } t, \text{ no burst}\}, \quad (5.5)$$

$$I_{\text{fix}}(t) = \mathbb{P}\{X_t \notin \{0, H\}\} = I(t) - D(t), \quad (5.6)$$

the last being the probability that the cell is still productively infected.

The first two satisfy the same Riccati equation,

$$\frac{dI}{dt} = \lambda(I - a)(I - b) = \mu - (\lambda + \mu + \delta)I + \lambda I^2, \quad (5.7)$$

differently initialised ($I(0) = 1, D(0) = 0$); its branching term I^2 comes from the two independent daughter lineages that a birth of the founder creates. Separating variables, with $w := e^{\theta t} \geq 1$, gives the triplet

$$\boxed{I(t) = \frac{aB + bAw}{B + Aw}, \quad D(t) = \frac{ab(w-1)}{aw-b}, \quad I_{\text{fix}}(t) = \frac{(a-b)^2 w}{(B + Aw)(aw-b)},} \quad (5.8)$$

with $I(\infty) = D(\infty) = b$, $I_{\text{fix}}(\infty) = 0$, $I_{\text{fix}}(0) = 1$ and $I'_{\text{fix}}(0) = -(\mu + \delta)$: at founding the productive state is lost only by the death or the catastrophe of the single founder. [Corollary 5.5](#) confirms the third formula independently.

The remaining closed forms are derived in the chapter on the birth–death–catastrophe process. The hazard of bursting is δ times the current load, so $I' = -\delta J$, and

$$J(t) = \mathbb{E}[X_t] = -\frac{1}{\delta} \frac{dI}{dt} = \frac{(a-b)^2 w}{(B + Aw)^2} = -\frac{\lambda}{\delta} (I - a)(I - b), \quad (5.9)$$

$$K(t) = \mathbb{E}[X_t^2] = \left[1 + \frac{2\lambda}{\delta}(1 - I)\right]J = \frac{2\lambda}{\delta}(\kappa - I)J, \quad \kappa = 1 + \frac{\delta}{2\lambda}, \quad (5.10)$$

$$V(t) = \mathbb{E}[W_t], \quad V'(t) = \delta K(t), \quad V(t) = (1 - I)\left[1 + \frac{\lambda}{\delta}(1 - I)\right], \quad (5.11)$$

$$V_{\infty} = \frac{a(1-b)}{a-1} = \frac{aB}{A}, \quad \mathbb{E}[\mathcal{K} \mid \text{burst}] = \frac{V_{\infty}}{1-b} = \frac{a}{a-1}. \quad (5.12)$$

One small point carries large consequences. $V_{\infty} = \mathbb{E}[\mathcal{K} \mathbf{1}\{\text{burst}\}]$ *includes* the realisations that cleared internally and released nothing, a mass b at 0, whereas the conditional mean removes them; [section 5.8](#) takes the distinction apart.

The assay predictions of [section 5.12](#) need the second moment of the release as well. The chapter on the birth–death–catastrophe process obtains it from the moment hierarchy, eliminating $\mathbb{E}[X_t^3]$ through the equation for K :

$$\boxed{\mathbb{E}[W_t^2] = \frac{2(\lambda - \mu)}{\delta} V - \frac{\lambda + \mu}{\delta} I - K + \frac{\lambda + \mu}{\delta} + 1,} \quad (5.13)$$

with $\text{Var}(W_t) = \mathbb{E}[W_t^2] - V^2$. At $\mu = 0$ and $t \rightarrow \infty$ it reduces to $2\lambda^2/\delta^2 + 3\lambda/\delta + 1$, the second moment of the geometric burst-size law of [section 5.6.1](#) at success probability $s = \delta/(\lambda + \delta)$.

Every formula above is a moment, and every one is quoted rather than derived here. What produces them has not been written down yet.

5.3 The generating function

Everything in this chapter is read off one object. The PGF of the load,

$$G(z, t) = \sum_{n \geq 0} p_n(t) z^n, \quad p_n(t) = \mathbb{P}\{X_t = n\}, \quad (5.14)$$

packs the entire state distribution into a single function of two variables and satisfies a first-order partial differential equation that can be solved exactly. We derive that equation, solve it, and record what the two rupture conventions do to the answer.

One feature of the catastrophe convention should be named before it is used. Ruptured realisations are absorbed into the separate state H and no longer contribute to the count, so $G(1, t) = I(t) \leq 1$: the PGF is *defective*, and its mass decays as cells burst. Under killing the rupture transition lands in state 0 and is counted, so $G(1, t) = 1$ for all t .

5.3.1 From the master equations to a PDE

The forward Kolmogorov equations for the killing convention are

$$\frac{dp_i}{dt} = \lambda(i-1)p_{i-1} + \mu(i+1)p_{i+1} - (\lambda + \mu + \delta)ip_i \quad (i > 0), \quad \frac{dp_0}{dt} = \mu p_1 + \delta \sum_{j \geq 1} jp_j. \quad (5.15)$$

The summation collects every rupture transition — from any positive state a collapse to 0 can occur — and is itself the mean $\mathbb{E}[X_t] = J(t)$. Under catastrophe that term is absent, since rupture leaves the count altogether; the first equation then holds for all $i \geq 0$ with $p_{-1} = 0$, and at $i = 0$ reads

$$\frac{dp_0}{dt} = \mu p_1, \quad (5.16)$$

internal extinction by death still feeding state 0.

Multiply the i th equation by z^i and sum over i . Shifting indices in the three sums and using $\sum_{i \geq 1} ip_i z^{i-1} = \frac{\partial}{\partial z} \sum_{i \geq 1} p_i z^i = \frac{\partial G}{\partial z}$ collapses the infinite system into one equation. Under killing,

$$\frac{\partial G}{\partial t} = \delta J(t) + (\lambda z^2 - (\lambda + \mu + \delta)z + \mu) \frac{\partial G}{\partial z}, \quad (5.17)$$

and dropping the inhomogeneous term — no rupture flux enters any state of the count under catastrophe — leaves

$$\frac{\partial G}{\partial t} = (\lambda z^2 - (\lambda + \mu + \delta)z + \mu) \frac{\partial G}{\partial z}. \quad (5.18)$$

5.3.2 Solving by characteristics

(5.18) is first order and linear, so it reduces to a family of ordinary differential equations along characteristic curves. The generic construction — the change of variables $(z, t) \rightarrow (s, \xi)$, the admissible initial curves, and the transport equations $\frac{dG}{d\xi} = 0$ and $\frac{dt}{d\xi} = -1$ — is set out in [section 1.3.3](#), with the generic system at (1.80); we take it from there and record only what is specific to this process. With the initial curve $z(s, 0) = s$, $t(s, 0) = 0$, transport gives $t = -\xi$ and G constant on curves of fixed s , so that the initial condition $G(z, 0) = z$ — a single founder, $p_1(0) = 1$ — fixes $G(s, \xi) = s$.

That leaves one equation, and it is the one that is not generic:

$$\frac{dz}{d\xi} = \lambda z^2 - (\lambda + \mu + \delta)z + \mu. \quad (5.19)$$

This is the Riccati equation (5.7) of the no-rupture probability, in a different variable and with a different initial condition: $I(0) = 1$ there, $z(s, 0) = s$ here. Its solution therefore reproduces the formula for I with s wherever a 1 stood,

$$z = \frac{a(s-b) - b(s-a)e^{\theta\xi}}{(s-b) - (s-a)e^{\theta\xi}}, \quad (5.20)$$

now in ξ rather than t . Since $G(z, t) = s(z, t)$, one step finishes the calculation: rearrange for s and put $-t$ for ξ . The map is an involution — s and z exchange places and nothing else moves — so

$$s = \frac{a(z-b) - b(z-a)w}{(z-b) - (z-a)w}, \quad w = e^{\theta t}, \quad (5.21)$$

and the generating function of the single-founder process is

$$G(z, t) = \frac{a(z-b) - b(z-a)w}{(z-b) - (z-a)w} = \frac{ab(1-w) - z(a-bw)}{(b-aw) - z(1-w)}. \quad (5.22)$$

The second form is the one to differentiate: its z -dependence is one linear factor over another, so repeated ∂_z returns a power over a power, and [section 5.4](#) reads the state probabilities off it.

5.3.3 Several founders

The solution extends to any initial load by the branching property. If the cell begins with k independent founders, the k lineages evolve independently until the shared catastrophe clock rings and the total count is their sum, so the PGF is the k th power of the single-founder one:

$$G_k(z, t) = (G(z, t))^k. \quad (5.23)$$

Under catastrophe this is defective, with $G_k(1, t) = I(t)^k$; under killing it remains proper. (5.23) is what sections 5.9 and 5.10 are built on, and it is where the independence assumption of section 5.1.1 does its work.

5.3.4 The two conventions differ by a function of time

The killing convention needs no second derivation. Its PDE (5.17) carries the inhomogeneous term $\delta J(t)$, which is $-\frac{dI}{dt}$; along a characteristic that term integrates to I alone, with no z in it. The conclusion is exact and immediate.

Proposition 5.3 (The two rupture conventions). *For every z and every t ,*

$$G_{\text{kill}}(z, t) = G_{\text{cat}}(z, t) + (1 - I(t)). \quad (5.24)$$

Proof. Along the characteristics of section 5.3.2 the killing system differs from the catastrophe system in its first equation alone, which becomes $\frac{dG}{d\xi} = -\delta J(-\xi) = \frac{d}{d\xi} I(-\xi)$, since $dt = -d\xi$ and $I' = -\delta J$. Hence $G(s, \xi) = -I(-\xi) + \phi(s)$, and $G(s, 0) = s$ with $I(0) = 1$ fixes $\phi(s) = 1 + s$, so $G(s, \xi) = 1 - I(-\xi) + s$. The remaining characteristic equation is (5.19) unchanged, giving (5.21) again; substituting yields

$$G_{\text{kill}}(z, t) = 1 - I(t) + \frac{ab(1 - w) - z(a - bw)}{(b - aw) - z(1 - w)}, \quad (5.25)$$

which is (5.22) plus $1 - I(t)$. □

Three consequences follow in a line each. The state probabilities agree, $p_n^{\text{kill}} = p_n^{\text{cat}}$ for every $n \geq 1$. The difference is confined to the empty state, $p_0^{\text{kill}} = p_0^{\text{cat}} + (1 - I)$. Every moment of X_t restricted to $n \geq 1$ is therefore convention-independent, so nothing in the rest of this chapter would change had killing been chosen instead. One reason to prefer catastrophe nonetheless survives the equivalence: it keeps $G(1, t) = I(t)$, so that the defect of the generating function is itself the biological observable, the probability that the cell has not yet ruptured. That is why the convention was chosen, and why the parallel track need not be carried further.

With the convention settled, the generating function can be read for what it contains. Extracting its coefficients is mechanical; what comes out of them is not.

5.4 The load is geometric at every time

The generating function (5.22) is a ratio of two linear functions of z , and that fact alone decides the shape of the whole distribution. Extracting the coefficients returns not an unstructured sequence but a geometric law, at every time, with a ratio that moves. Naming that ratio is the single most useful thing this chapter does, because every result after this section is a statement about where it goes.

Coefficients come out of a PGF by repeated differentiation at the origin, $p_n = \frac{1}{n!} \partial_z^n G|_{z=0}$; the construction is given in section 1.D at (1.132). Applied to (5.22), the derivatives close on themselves,

$$\frac{\partial^n G}{\partial z^n} = n! (a - b)^2 w \frac{(1 - w)^{n-1}}{(b - aw - (1 - w)z)^{n+1}}, \quad (5.26)$$

so that setting $z = 0$ and dividing by $n!$ gives, for $n \geq 1$,

$$p_n(t) = \left(\frac{a-b}{b-aw} \right)^2 w \left(\frac{1-w}{b-aw} \right)^{n-1}. \quad (5.27)$$

The empty state is the only one that sees the convention. Under catastrophe $p_0 = G(0, t) = ab(1-w)/(b-aw)$, the probability of reaching 0 by the death of every individual; under killing it carries the extra $1 - I(t)$ of [proposition 5.3](#), and nothing else changes.

5.4.1 The sliding ratio

There is more structure in (5.27) than it first shows. The entire dependence on n sits in one factor. Write

$$P(t) := \frac{1-w}{b-aw}, \quad w = e^{\theta t}, \quad (5.28)$$

and the state probabilities become $p_n(t) = p_1(t)P(t)^{n-1}$.

Theorem 5.4 (The load is geometric at every time). *For every $t > 0$, conditional on the load being non-zero, X_t is geometrically distributed on $\{1, 2, \dots\}$ with ratio $P(t)$:*

$$p_n(t) = p_1(t)P(t)^{n-1}, \quad n \geq 1. \quad (5.29)$$

The ratio increases monotonically in t , from

$$P(0) = 0 \quad \text{to} \quad P(\infty) = \frac{1}{a}. \quad (5.30)$$

Proof. Equation (5.29) is (5.27) read with the n -dependent factor named. For the limits, $w = 1$ at $t = 0$ gives $P(0) = 0$, and $w \rightarrow \infty$ gives $P \rightarrow 1/a$; monotonicity follows from $P'(t) = \lambda(a-b)^2 w / (b-aw)^2 > 0$, since $b-aw$ never vanishes for $t \geq 0$. $\square$

What P is, is worth saying plainly. It is the price in probability of one more particle: $p_{n+1}/p_n = P(t)$ for every $n \geq 1$, so each additional unit of load is the same factor less likely than the one below it, whatever that load happens to be. At founding $P = 0$ and the cell holds its single particle with certainty. Thereafter P rises, the geometric flattens, and the mean load conditional on non-fixation, $1/(1-P)$, rises with it — from 1 at founding towards $a/(a-1)$. That the rise never reverses is what [theorem 5.4](#) establishes, and it is what makes the endpoint well defined: a cell that has survived longer is spread over more states, never fewer.

So the transient laws form a one-parameter family of geometrics sliding towards a fixed limit, and P is the only thing in them that moves. [Figure 5.3](#) shows the slide: on a logarithmic scale each law is a straight line, and the lines rotate onto a limiting slope rather than deforming. The limit $1/a$ is where the next three sections live — [section 5.5](#) takes it, [section 5.6](#) integrates against it, [section 5.7](#) conditions on it. [Figure 5.2](#) is that sentence drawn.

5.4.2 Normalisation: the sum is I_{fix}

One consequence is immediate, and it checks the closed forms of [section 5.2](#) against a derivation that shares nothing with them.

Corollary 5.5 (The positive states carry the non-fixation probability). *For every $t \geq 0$,*

$$\sum_{n \geq 1} p_n(t) = \frac{(a-b)^2 w}{(B+Aw)(aw-b)} = I_{\text{fix}}(t). \quad (5.31)$$

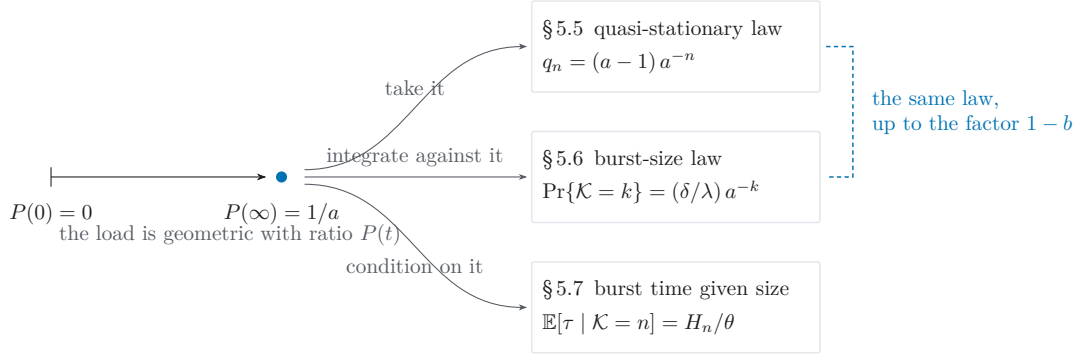


Figure 5.2: One endpoint, three readings. The ratio $P(t)$ of [theorem 5.4](#) slides from 0 at founding to $1/a$, and each of the next three sections applies a different operation to that endpoint: [section 5.5](#) takes it as the quasi-stationary ratio, [section 5.6](#) integrates the rupture hazard against it, and [section 5.7](#) conditions the rupture time on the size it produces. The first two return the same law, differing only by the eventual-burst probability $1 - b$; that coincidence is [theorem 5.10](#).

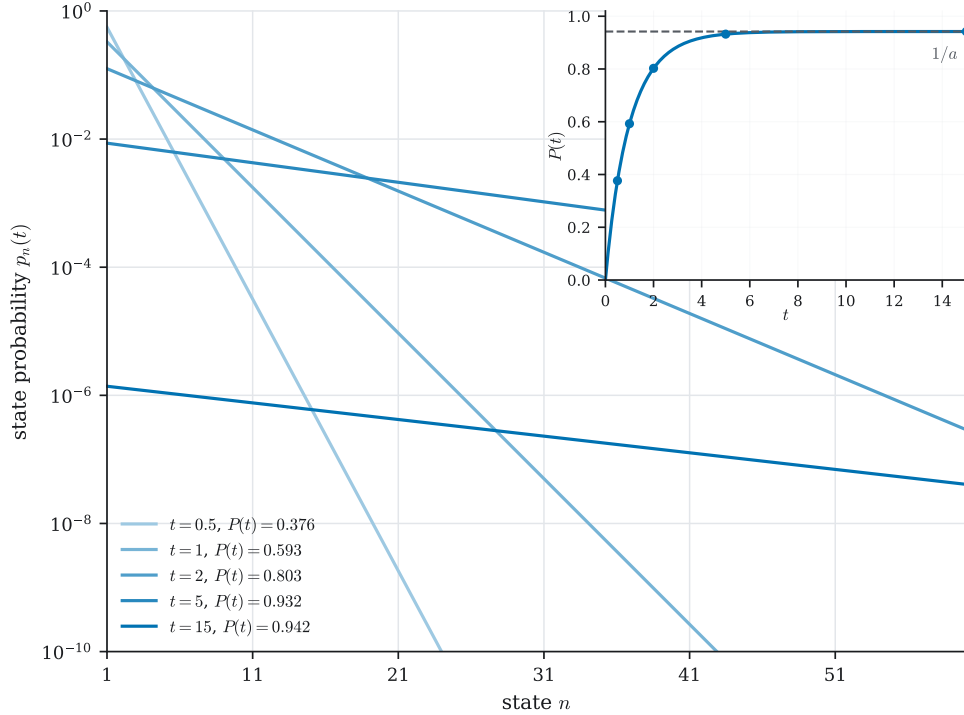


Figure 5.3: State probabilities $p_n(t)$ at successive times for $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$. On the logarithmic vertical scale each geometric law is a straight line whose ratio $P(t)$ increases towards $1/a$; the inset shows that approach directly. The curves fall in n and rise in t : the load spreads as P slides.

Proof. By [theorem 5.4](#) the sum is the geometric series $p_1(t)/(1 - P(t))$. Now

$$1 - P(t) = \frac{b - aw - 1 + w}{b - aw} = \frac{-(1 - b) - (a - 1)w}{b - aw} = \frac{B + Aw}{aw - b},$$

and dividing $p_1(t) = (a - b)^2 w / (b - aw)^2$ by this gives the middle expression, which is the closed form for I_{fix} in [\(5.8\)](#). $\square$

The sum of the positive state probabilities is exactly the probability of being neither ruptured nor extinct. The chapter on the birth–death–catastrophe process reached the same I_{fix} by decomposing absorption rather than by summing a distribution; that two such different routes agree is a check on both.

5.5 The quasi-stationary distribution

The BDC process is absorbed in the end, either at the rupture state H or by death at 0; the theory of what a process looks like before that happens is quasi-stationarity [\[7\]](#). A realisation which has, by chance, not yet reached fixation has had to hold a middle course. Fly too low and depletion to zero becomes more and more probable; fly high, a population booming in number, and the catastrophe clock has all the more to ring against. What survives hovers, on Daedalus’s instruction exactly: neither so low that the spray soaks the wings, nor so high that the sun reaches them.

A rogue process still going in defiance of the odds is described by the quasi-stationary distribution: a set of limiting conditional probabilities, each qualified by non-fixation. The general definition, the Yaglom limit and the classical existence theory are in [definition 1.7](#) and [section 1.3.2.6](#), resting on the classification of birth–death processes [\[8\]](#). Here the limit is quoted from nowhere: it is read off [theorem 5.4](#) by letting P reach its endpoint.

5.5.1 Conditioning on survival

Let $F(t)$ be the probability of fixation by time t , whether by death or by collapse. The decomposition of [section 5.2.4](#) settles it immediately: a realisation is fixed by t precisely when it is not productively infected, so

$$F(t) = 1 - I_{\text{fix}}(t) = 1 - I(t) + D(t), \quad (5.32)$$

which reduces to $1 - I(t)$ when $\mu = 0$. Conditional state probabilities decompose as

$$\mathbb{P}\{X_t = i\} = \mathbb{P}\{X_t = i \mid \text{fixation}\} F(t) + \mathbb{P}\{X_t = i \mid \text{non-fixation}\} (1 - F(t)),$$

and for $i \notin \{0, H\}$ the process has not been absorbed, so the first term vanishes and

$$\mathbb{P}\{X_t = i \mid \text{non-fixation}\} = \frac{p_i(t)}{1 - F(t)} = \frac{p_i(t)}{I_{\text{fix}}(t)}. \quad (5.33)$$

The quasi-stationary distribution is the limiting form of these:

$$q_i := \lim_{t \rightarrow \infty} \mathbb{P}\{X_t = i \mid X_t \notin \{0, H\}\} = \lim_{t \rightarrow \infty} \frac{p_i(t)}{I_{\text{fix}}(t)}, \quad i \geq 1. \quad (5.34)$$

5.5.2 The limit is geometric

Theorem 5.6 (Quasi-stationary distribution of the BDC process). *For all $\mu \geq 0$ the quasi-stationary distribution exists and is geometric on $\{1, 2, \dots\}$ with ratio $1/a$:*

$$q_n = \left(1 - \frac{1}{a}\right) \left(\frac{1}{a}\right)^{n-1} = (a - 1) a^{-n}, \quad n \geq 1. \quad (5.35)$$

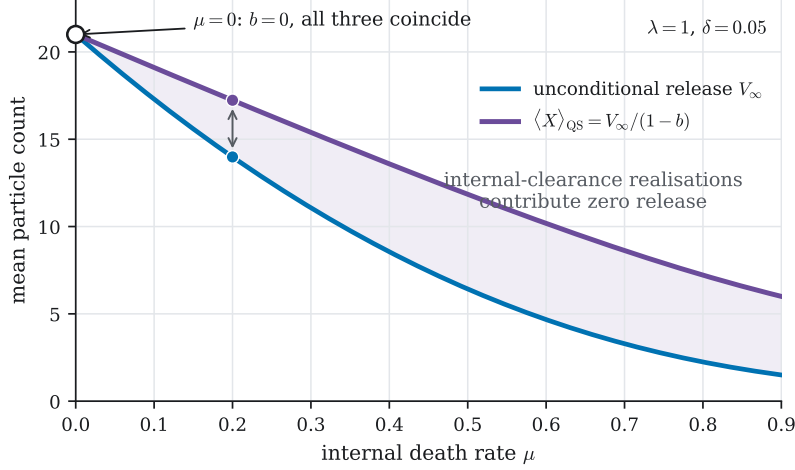


Figure 5.4: Quasi-stationary and release means across internal death rates at $\lambda = 1$ and $\delta = 0.05$. The quasi-stationary mean $\langle X \rangle_{\text{QS}} = a/(a-1)$ is identically the conditional burst mean $V_\infty/(1-b)$. At $\mu = 0$, $b = 0$ and both also equal V_∞ ; once $\mu > 0$, internally cleared cells contribute zero release and place V_∞ strictly below the other two; the arrow marks that gap, $b\langle X \rangle_{\text{QS}} = 3.25$, at the working point. In the $\mu = 0$ case the conditional mean and the cumulative release share a limit, $\lim_t J(t)/I(t) = \lim_t V(t) = 1 + \lambda/\delta$, which diverges as δ shrinks.

Proof. By (5.33) and corollary 5.5, the conditional law at time t is

$$\mathbb{P}\{X_t = n \mid X_t \notin \{0, H\}\} = \frac{p_n(t)}{\sum_{m \geq 1} p_m(t)} = (1 - P(t)) P(t)^{n-1}, \quad (5.36)$$

for every t : no limit has been taken, and the conditional law is already geometric. By theorem 5.4, $P(t) \uparrow 1/a$, so each coordinate converges and the limit is (5.35). $\square$

The theorem is a corollary of theorem 5.4 in substance as well as in proof: the quasi-stationary law is not a new object but the endpoint of a family that was geometric all along. Its moments follow immediately:

$$\langle X \rangle_{\text{QS}} = \frac{a}{a-1} = \frac{a}{A}, \quad \langle X^2 \rangle_{\text{QS}} = \frac{a(a+1)}{(a-1)^2}, \quad \text{Var}(X)_{\text{QS}} = \frac{a}{(a-1)^2} = \frac{a}{A^2}. \quad (5.37)$$

The quasi-stationary mean is exactly the conditional mean burst size $\mathbb{E}[K \mid \text{burst}]$ of (5.12), and at $\mu = 0$ both also equal V_∞ . Once $\mu > 0$ it is V_∞ that falls away from the other two: at $(1, 0.2, 0.05)$ it is 13.99 against $\langle X \rangle_{\text{QS}} = 17.23$, because V_∞ averages in the cells that cleared internally and released nothing. Figure 5.4 traces all three across μ . That the first two agree is not a coincidence of first moments: section 5.6 shows the entire burst-size distribution *is* the quasi-stationary distribution.

5.5.3 The decay rate, and what it validates

Quasi-stationarity has a spectral reading, and carrying it out closes a question left open in the mathematical introduction. Restricting the generator to the positive states gives the killed subgenerator $\mathcal{Q}$, with $\mathcal{Q}_{i,i+1} = \lambda i$, $\mathcal{Q}_{i,i-1} = \mu i$ for $i \geq 2$, and $\mathcal{Q}_{i,i} = -(\lambda + \mu + \delta)i$; its missing row mass is the instantaneous absorption rate, δi for $i \geq 2$ and $\mu + \delta$ for $i = 1$. Writing T_{fix} for the first of internal extinction and rupture and summing the positive-state forward equations,

$$\frac{d}{dt} \mathbb{P}\{T_{\text{fix}} > t\} = -\mu \mathbb{P}\{X_t = 1\} - \sum_{i \geq 1} \delta i \mathbb{P}\{X_t = i\}, \quad (5.38)$$

all probabilities on the right understood before absorption. Ordinary death drains survival mass from state 1 only; catastrophe drains it from every positive state.

Proposition 5.7 (Quasi-stationary decay rate). *If ν is a quasi-stationary distribution on the positive states and the sums below are finite, then for some $\vartheta > 0$*

$$\nu \mathcal{Q} = -\vartheta \nu, \quad \mathbb{P}_\nu\{T_{\text{fix}} > t\} = e^{-\vartheta t}, \quad \vartheta = \mu \nu_1 + \sum_{i \geq 1} \delta i \nu_i. \quad (5.39)$$

Proof. Quasi-stationarity writes the unnormalised positive-state law at time t as $\alpha(t)\nu$; the semigroup property forces $\alpha(s+t) = \alpha(s)\alpha(t)$, and right-continuity gives $\alpha(t) = e^{-\vartheta t}$. Differentiating at zero yields the left-eigenvector relation, and summing its coordinates against (5.38) identifies the decay rate. $\square$

The state-dependence matters. Were the catastrophe rate a constant, independent of the load, its clock would run independently of the birth–death process, the killed semigroup would factorise into that clock’s exponential times its birth–death counterpart, and the extra clock would shift the decay rate by that constant while leaving the conditional law untouched. A rate proportional to the load weights paths by their accumulated exposure and generally changes the quasi-stationary law itself. It is therefore no formality that a quasi-stationary distribution survives conditioning on *neither* extinction nor rupture; the mathematical introduction leaves that open, and theorem 5.6 settles it in closed form.

Substituting that law, $\nu_n = (a-1)a^{-n}$, into (5.39) gives a check on everything together. Using $\nu_1 = (a-1)/a$, $\sum_i i \nu_i = a/(a-1)$, and then $\mu = \lambda ab$ and $\delta = \lambda(a-1)(1-b)$ from (5.4),

$$\vartheta = \mu \frac{a-1}{a} + \delta \frac{a}{a-1} = \lambda ab \frac{a-1}{a} + \lambda(a-1)(1-b) \frac{a}{a-1} = \lambda b(a-1) + \lambda a(1-b) = \lambda(a-b) = \theta. \quad (5.40)$$

The eigenvalue of the killed subgenerator is the time constant that has appeared in every formula since (5.3). One line validates the quasi-stationary law, the decay-rate formula and $AB = \delta/\lambda$ together.

5.5.4 Mean productive lifetime

One integral of I_{fix} is a result in its own right, and is also the first item of the consignment named in section 5.1.3. The expected duration of productive infection is

$$\mathbb{E}[T_{\text{prod}}] = \int_0^\infty I_{\text{fix}}(t) dt, \quad (5.41)$$

since $I_{\text{fix}}(t) = \mathbb{P}\{T_{\text{prod}} > t\}$.

Proposition 5.8 (Mean productive lifetime).

$$\mathbb{E}[T_{\text{prod}}] = \frac{1}{\lambda} \log\left(\frac{a}{a-1}\right) = \frac{1}{\lambda} \log\langle X \rangle_{\text{QS}}. \quad (5.42)$$

Proof. Write $I_{\text{fix}} = (I-b) - (D-b)$. From (5.8),

$$I(t) - b = \frac{B(a-b)}{B + Aw}, \quad D(t) - b = -\frac{b(a-b)}{aw - b},$$

so that, with $w = e^{\theta t}$ and $dt = dw/(\theta w)$,

$$\int_0^\infty I_{\text{fix}} dt = \frac{a-b}{\theta} \int_1^\infty \left(\frac{B}{w(B + Aw)} + \frac{b}{w(aw - b)} \right) dw.$$

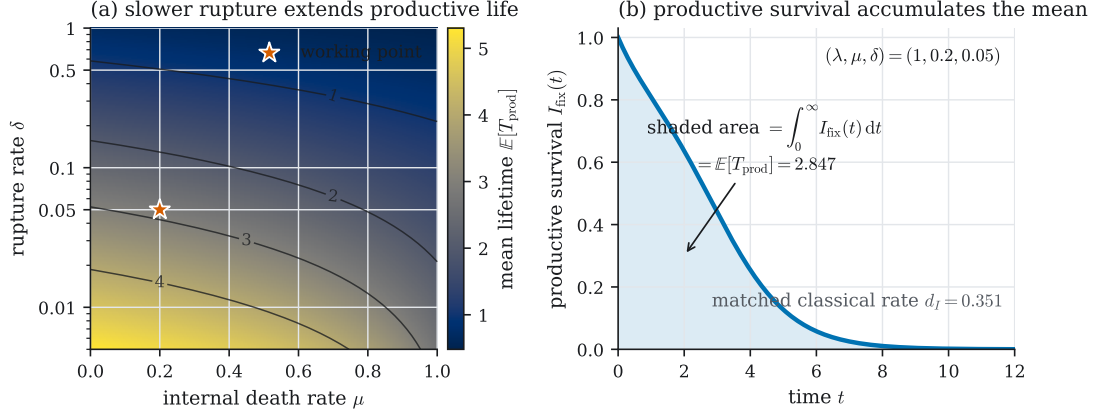


Figure 5.5: Mean productive lifetime $\mathbb{E}[T_{\text{prod}}] = \lambda^{-1} \log(a/(a-1))$. (a) Its dependence on $\mu \in [0, 1]$ and $\delta \in [0.005, 1]$ at $\lambda = 1$; contours give lifetime values and the star marks the working point. (b) The productive-survival curve $I_{\text{fix}}(t)$ at $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, with its integral $\mathbb{E}[T_{\text{prod}}] = 2.847$ shaded. The matched classical death rate for the chapter on population dynamics is $d_{\mathcal{I}} = 1/\mathbb{E}[T_{\text{prod}}] = 0.351$.

By partial fractions $\frac{B}{w(B+Aw)} = \frac{1}{w} - \frac{A}{B+Aw}$ and $\frac{b}{w(aw-b)} = -\frac{1}{w} + \frac{a}{aw-b}$, hence

$$\begin{aligned} \int_1^\infty \frac{B dw}{w(B+Aw)} &= \left[\log \frac{w}{B+Aw} \right]_1^\infty = \log \frac{a-b}{a-1}, \\ \int_1^\infty \frac{b dw}{w(aw-b)} &= \left[\log \frac{aw-b}{w} \right]_1^\infty = \log \frac{a}{a-b}. \end{aligned}$$

Summing, and using $\theta = \lambda(a-b)$,

$$\int_0^\infty I_{\text{fix}} dt = \frac{a-b}{\lambda(a-b)} \left(\log \frac{a-b}{a-1} + \log \frac{a}{a-b} \right) = \frac{1}{\lambda} \log \frac{a}{a-1}. \quad \square$$

When $\mu = 0$ this is $\lambda^{-1} \log(1+\lambda/\delta)$. The form of the answer interprets itself. Here $\log \langle X \rangle_{\text{QS}}/\lambda$ is the time that exponential growth at rate λ would take to reach the quasi-stationary level — the productive lifetime is precisely the time needed to climb to the height at which the process can hover. Figure 5.5 maps it across the rate plane; at the working parameters $\mathbb{E}[T_{\text{prod}}] = 2.847$, implying a matched constant-rate death rate $d_{\mathcal{I}} = 0.351$.

That is the cell while it lasts. What it releases when it stops is a separate question, and the law answering it has already appeared in this section.

5.6 Burst size, and why it is the quasi-stationary law

A cell presents two quantities to the outside world: when it ruptures, and how much it releases. The second is settled first, because it is the law the previous section has just computed.

5.6.1 The burst-size law

The burst size $\mathcal{K}$ is the load immediately before rupture. For a burst in state k the process must be at k and then rupture from there, which happens at rate δk , so

$$\mathbb{P}\{\mathcal{K} = k\} = \delta k \int_0^\infty p_k(t) dt. \quad (5.43)$$

A discretisation makes this plain: over an interval of width Δt in which the process is at k , the probability that the burst occurs there is $\mathbb{P}\{X_t = k\} \delta k \Delta t$, and summing over intervals as $\Delta t \rightarrow 0$ gives (5.43).

The integral is elementary once the sliding ratio is in view. With P as in (5.28),

$$\frac{d}{dt} \left[\frac{P(t)^k}{k\lambda} \right] = \frac{1}{\lambda} P(t)^{k-1} P'(t) = \left(\frac{a-b}{b-aw} \right)^2 w P(t)^{k-1} = p_k(t), \quad (5.44)$$

as one verifies from $P'(t) = \lambda(a-b)^2 w / (b-aw)^2$. Since $P(0) = 0$ and $P(\infty) = 1/a$,

$$\int_0^\infty p_k(t) dt = \frac{1}{k\lambda} \left(\frac{1}{a} \right)^k, \quad (5.45)$$

and (5.43) gives the law.

Theorem 5.9 (Burst-size distribution). *Unconditionally, including the mass b of realisations that never burst,*

$$\mathbb{P}\{\mathcal{K} = k\} = \frac{\delta}{\lambda} a^{-k}, \quad k \geq 1; \quad (5.46)$$

conditioned on the burst occurring,

$$\mathbb{P}\{\mathcal{K} = k \mid \text{burst}\} = (a-1) a^{-k}, \quad k \geq 1. \quad (5.47)$$

Proof. The unconditional form is the computation above. For the conditional form, divide (5.46) by the eventual-burst probability $1-b$:

$$\mathbb{P}\{\mathcal{K} = k \mid \text{burst}\} = \frac{(\delta/\lambda) a^{-k}}{1-b} = \frac{AB}{B} a^{-k} = (a-1) a^{-k},$$

using $\delta/\lambda = AB$ from (5.4). □

Two consistency checks close the loop. Normalisation:

$$\sum_{k \geq 1} \frac{\delta}{\lambda} a^{-k} = \frac{\delta}{\lambda} \frac{1}{a-1} = \frac{AB}{A} = B = 1-b,$$

matching the eventual-burst probability. And the first moment:

$$\sum_{k \geq 1} k \frac{\delta}{\lambda} a^{-k} = \frac{\delta}{\lambda} \frac{a}{(a-1)^2} = \frac{aB}{A} = V_\infty,$$

recovering (5.12) — which validates the burst-size law, the formula for V_∞ and the identity (5.4) at once.

5.6.2 Why the two laws agree

Set (5.47) beside (5.35). They are the same law.

Theorem 5.10 (The burst size is the quasi-stationary law). *The burst-size distribution conditioned on bursting is exactly the quasi-stationary distribution of the load: both are geometric on $\{1, 2, \dots\}$ with ratio $1/a$.*

That the algebra agrees is not by itself an explanation, because the two constructions share no step. On one side, condition the load on having avoided both absorbing states and let $t \rightarrow \infty$: a limit along a single trajectory, in which nothing is averaged and no rupture occurs. On the other, weight each state by its rupture hazard δk and integrate over all rupture times: an average across trajectories, in which every realisation contributes at the instant it dies and never afterwards. Nothing forces a conditional limit and an occupation integral to agree, and for a killed process whose hazard was not linear in the load they would not. Figure 5.6 sets the two side by side.

Proof. Read (5.44) as a change of clock rather than as an antiderivative. Its $k = 1$ case is

$$p_1(t) = \frac{P'(t)}{\lambda}, \quad (5.48)$$

and with $\sigma = P(t)$, which runs monotonically from 0 to $1/a$ by [theorem 5.4](#), the occupation density of state k becomes a perfect differential:

$$p_k(t) dt = \frac{1}{\lambda} \sigma^{k-1} d\sigma. \quad (5.49)$$

Hence

$$\mathbb{P}\{\mathcal{K} = k\} = \delta k \int_0^\infty p_k dt = \frac{\delta}{\lambda} \int_0^{1/a} k \sigma^{k-1} d\sigma = \frac{\delta}{\lambda} [\sigma^k]_0^{1/a} = \frac{\delta}{\lambda} a^{-k}, \quad (5.50)$$

which normalised on the bursting event is $(a-1)a^{-k}$, the law of (5.35). $\square$

The proof says more than the statement. The size-bias weight k is exactly the Jacobian that turns the occupation density of state k into a perfect differential, and integrating a perfect differential returns only its endpoints. The burst-size law therefore depends on the whole trajectory *only* through the endpoint $\sigma_\infty = 1/a$ that P reaches, which is by definition the quasi-stationary ratio. That is why the two laws coincide: not a coincidence, and not a Yaglom limit, but size-biasing telescoping the time integral onto the terminal value of P , the process's own geometric ratio.

Remark 5.11 (What the argument needs). Nothing in the proof used the birth–death structure. The identity holds for any killed process whose transient conditional law is geometric in a single sliding ratio $P(t)$ satisfying $p_1 = P'/\lambda$, with a killing hazard linear in the load. Whether that criterion *characterises* the class of processes for which burst size equals the quasi-stationary law is not settled here; see [section 5.12.4](#).

5.6.3 Size-biasing and the size of a late burst

The size-biasing that drives the telescoping has a reading of its own: it is what an experiment sees when it times its observations. A cell bursts at rate δX_t , so given a burst at time t its size is a size-biased draw from $p.(t)$:

$$\mathbb{P}\{\mathcal{K} = k \mid \tau = t\} = \frac{k p_k(t)}{J(t)}, \quad \mathbb{E}[\mathcal{K} \mid \tau = t] = \frac{K(t)}{J(t)} = 1 + \frac{2\lambda}{\delta} (1 - I(t)). \quad (5.51)$$

Letting $t \rightarrow \infty$,

$$\mathbb{E}[\mathcal{K} \mid \tau = t] \xrightarrow[t \rightarrow \infty]{} 1 + \frac{2\lambda}{\delta} (1 - b) = \frac{\langle X^2 \rangle_{\text{QS}}}{\langle X \rangle_{\text{QS}}} = \frac{a+1}{a-1}, \quad (5.52)$$

the second equality from (5.37). A late burst is therefore larger than an average one by the factor

$$\frac{(a+1)/(a-1)}{a/(a-1)} = \frac{a+1}{a},$$

which is 1.94 at the working parameters but only 1.38 at the *B. anthracis* parameters of [section 5.12.2](#): the often-quoted factor of two is the $a \rightarrow 1$ limit, not a general fact. [Figure 5.7](#) shows the rise.

The mechanism is a selection effect on the geometric law, not on individual cells. Conditioning on a late rupture makes no cell grow; it discards the cells that ruptured early, and early ruptures come disproportionately from small loads, the hazard δk being proportional to the load. What survives is a sample already weighted towards large loads, which (5.51) then weights once more at the moment of rupture. [Figure 5.8](#) makes the tilt visible.

The size of a burst is now fully described. Its timing is not, and the two are not independent — which is what makes the remaining conditioning worth doing.

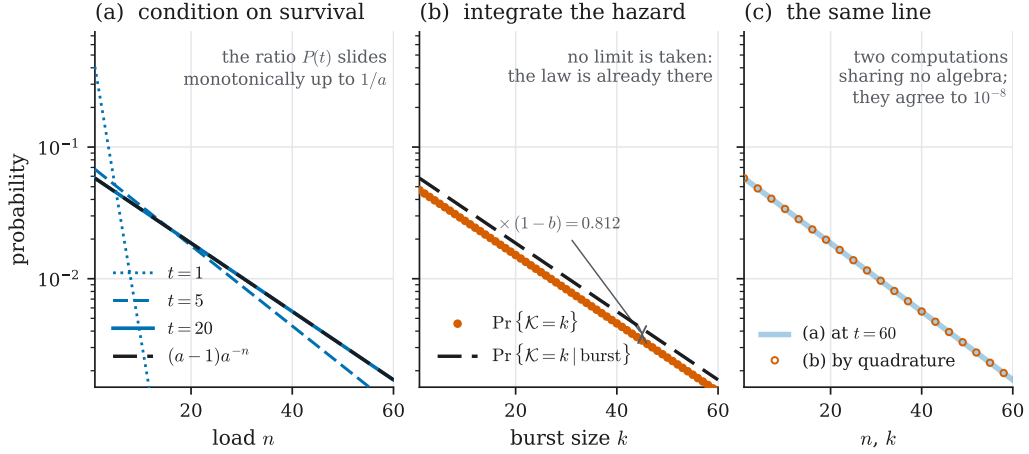


Figure 5.6: Two mechanisms, one law. (a) The load conditioned on being neither ruptured nor extinct, $p_n(t)/I_{\text{fix}}(t)$, at $t = 1, 5, 20$ from (5.8) and (5.27), with the quasi-stationary law $(a-1)a^{-n}$ dashed; on the logarithmic scale each transient law is a straight line whose slope rotates onto the limiting slope as $P(t)$ slides up to $1/a$. (b) The burst-size law $(\delta/\lambda)a^{-k}$ of (5.46) (markers) and the same law conditioned on the burst occurring (dashed) — the identical line, reached without taking a limit at all; the two series differ by the constant factor $1-b = 0.812$. The first panel arrives by conditioning on survival along a single trajectory, the second by weighting each state by its rupture hazard and integrating across trajectories. (c) The two routes on one axis, each computed on its own terms: the conditional law of (a) evaluated at $t = 60$, against the occupation integral of (b) evaluated by numerical quadrature rather than in closed form. They agree to 10^{-8} , which is the accuracy of the quadrature grid and not of the identity — theorem 5.10 makes the two exactly equal. Parameters $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, so $a = 1.0616$ and $b = 0.1884$; closed forms throughout except where panel (c) integrates numerically, and no simulation anywhere.

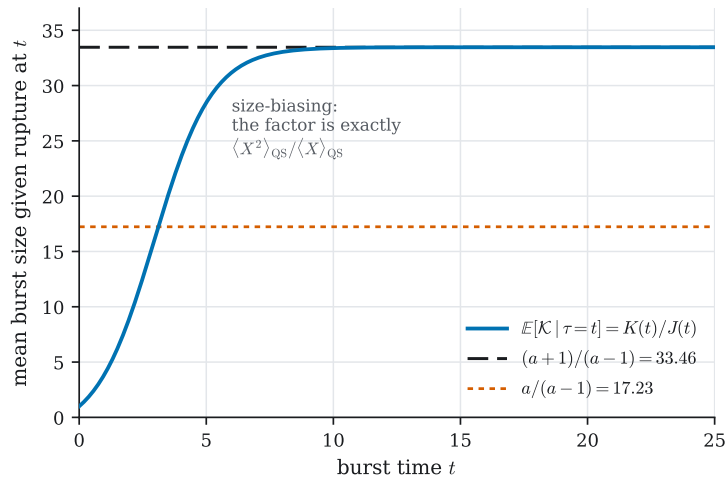


Figure 5.7: The size of a late burst at $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$. The conditional mean $\mathbb{E}[\mathcal{K} | \tau = t] = K(t)/J(t)$ of (5.51) rises from 1 at founding to $(a+1)/(a-1) = 33.46$; the quasi-stationary mean $a/(a-1) = 17.23$ is marked for contrast, their ratio being $(a+1)/a = 1.94$.

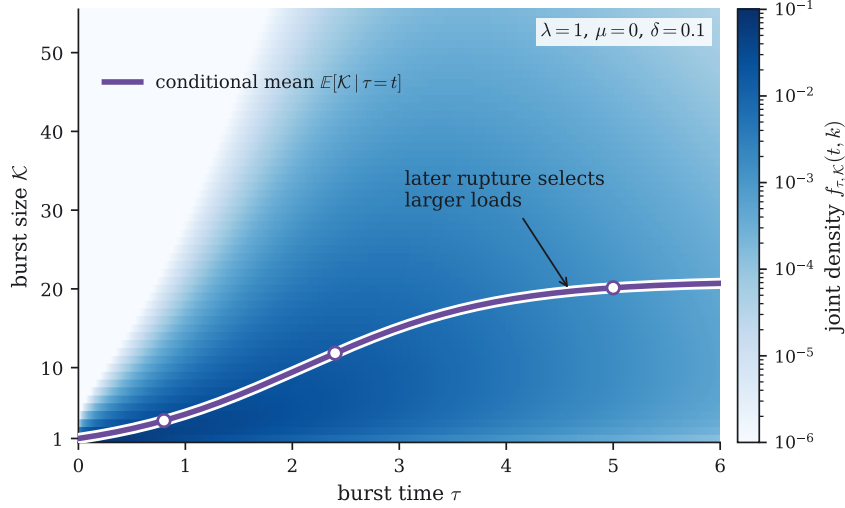


Figure 5.8: Joint density, with respect to time and counting measure, of burst time τ and burst size K for $(\lambda, \mu, \delta) = (1, 0, 0.1)$. The overlay is the conditional mean $\mathbb{E}[K | \tau = t] = K(t)/J(t)$, rising from 1 towards $1 + 2\lambda/\delta = 21$; the upward tilt of the mass is size-biasing made visible.

5.7 Burst time

Knowing what a burst is makes it possible to say when one happens. The marginal law of the rupture time is a one-line consequence of the survival function derived in the chapter on the birth–death–catastrophe process, and is taken first. The conditional law is the interesting one, and it runs the other way from [section 5.6.3](#): there the burst size was conditioned on the burst time, here the burst time on the burst size, which is what an experiment measures when it records both. By [theorem 5.10](#) that size is a draw from the law at the endpoint of P , the last of the three readings of that endpoint promised in [section 5.4](#).

5.7.1 The marginal law, and its defect

Since $I(t) = \mathbb{P}\{\tau > t\}$, the density of the burst time is

$$\varphi(t) := -I'(t) = \delta J(t), \quad (5.53)$$

using $I' = -\delta J$ from [section 5.2.4](#): the rupture flux at time t is the catastrophe rate times the expected load. The density is *defective*, because $I(t) \rightarrow b$ rather than 0:

$$\int_0^\infty \varphi(t) dt = 1 - b, \quad (5.54)$$

the remaining mass b sitting at $\tau = \infty$ — the realisations that clear internally without ever bursting. Conditioned on eventual burst the density is $\varphi(t)/(1 - b)$. [Figure 5.9](#) shows both, with the missing mass marked.

5.7.2 Burst time given burst size, without death

When $\mu = 0$ the conditional law of τ given K is completely explicit. Write $\theta = \lambda + \delta$ for the rate combination governing this case — it is $\lambda(a - b)$ at $b = 0$ — and $H_n = \sum_{j=1}^n 1/j$ for the n th harmonic number. Fixing n and reading $p_n(t)$ as a function of t , the state probability rises from zero, peaks, and decays; differentiating $\log p_n$ places the maximum at

$$t_n = \frac{\log n}{\theta}, \quad (5.55)$$

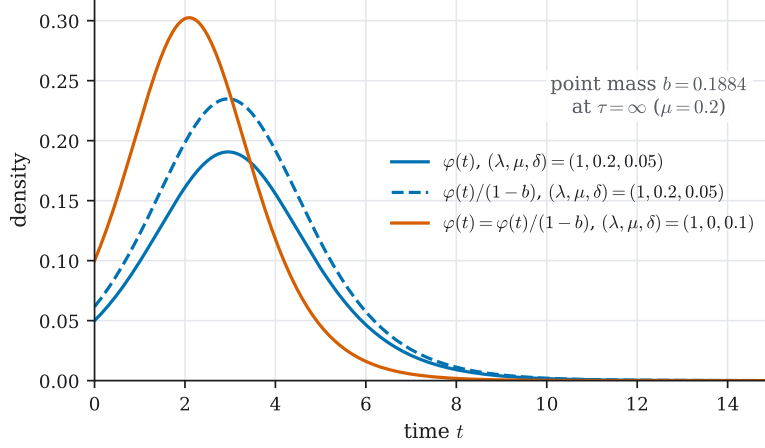


Figure 5.9: Defective burst-time density $\varphi(t) = \delta J(t)$ and its conditional counterpart $\varphi(t)/(1-b)$. For $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ the finite-time density has total mass $1-b$ and the remaining mass $b = 0.1884$ lies at $\tau = \infty$; for $(1, 0, 0.1)$, $b = 0$ and the two densities coincide. The gap between the curves is the whole content of the defect.

so a burst of size n is most likely to have occurred after about $(\log n)/\theta$ of growth. The exact conditional mean is a harmonic number.

Proposition 5.12 (Conditional rupture time, $\mu = 0$). *With $\mu = 0$ and $\theta = \lambda + \delta$,*

$$\mathbb{E}[\tau \mid \mathcal{K} = n] = \frac{1}{\theta} \sum_{k=1}^n \frac{1}{k} = \frac{H_n}{\theta}, \quad n \geq 1. \quad (5.56)$$

Proof. When $\mu = 0$ the state probabilities (5.27) reduce, with $r = \lambda/\delta$ and $s = \theta t$, to

$$p_n(t) = e^{-\theta t} \left(\frac{1 - e^{-\theta t}}{1 + 1/r} \right)^{n-1},$$

and the burst-size law (5.46) to $\mathbb{P}\{\mathcal{K} = n\} = (1/r)(r/(r+1))^n$. Writing

$$L_1(n) = \int_0^\infty e^{-s} (1 - e^{-s})^{n-1} ds, \quad L_2(n) = \int_0^\infty s e^{-s} (1 - e^{-s})^{n-1} ds,$$

the substitution $s = \theta t$ gives $\mathbb{E}[\tau \mid \mathcal{K} = n] = L_2(n)/(\theta L_1(n))$. The first integral is $L_1(n) = 1/n$, by the substitution $u = 1 - e^{-s}$. For the second, split off one factor of $(1 - e^{-s})$ to write $L_2(n+1) = L_2(n) - \int_0^\infty s e^{-2s} (1 - e^{-s})^{n-1} ds$, and integrate by parts in the remaining integral; the result is the recurrence

$$(n+1)L_2(n+1) = nL_2(n) + \frac{1}{n+1}, \quad L_2(1) = 1,$$

whose solution is $L_2(n) = H_n/n$. The quotient is (5.56). $\square$

For large n the harmonic sum gives

$$\mathbb{E}[\tau \mid \mathcal{K} = n] \simeq \frac{\log n + \gamma}{\theta} \quad (n \gg 1), \quad (5.57)$$

with $\gamma = 0.5772\dots$ Euler's constant: conditional rupture time grows logarithmically with burst size, so a burst ten times larger is late by a fixed increment rather than by a factor. Figure 5.10

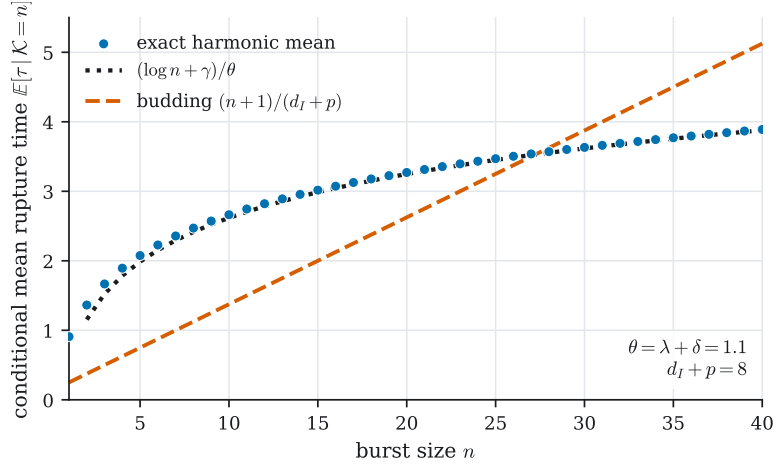


Figure 5.10: Conditional mean rupture time given burst size n when $\mu = 0$, with $\lambda = 1$, $\delta = 0.1$ and hence $\theta = 1.1$. Exact harmonic means $\theta^{-1}H_n$ from (5.56) are shown with the large- n approximation $\theta^{-1}(\log n + \gamma)$ and the linear budding comparator $(n+1)/(d_I + p)$ at $d_I + p = 8$. The two grow at different orders: logarithmically for bursting, linearly for budding.

sets the exact means against that approximation and against a budding comparator, whose conditional mean is linear in n .

The full conditional law has an equally clean form. The density of τ at t given $\mathcal{K} = n$ is proportional to $p_n(t)$, so

$$\mathbb{P}\{\tau < t \mid \mathcal{K} = n\} = \frac{\delta n \int_0^t p_n(u) du}{\mathbb{P}\{\mathcal{K} = n\}} = (1 - e^{-\theta t})^n, \quad (5.58)$$

which is the distribution function of the maximum of n independent exponentials of rate θ . Centring at t_n and scaling by $1/\theta$, with $T = \theta(t - t_n)$,

$$\mathbb{P}\{\tau < t \mid \mathcal{K} = n\} \simeq \exp(-ne^{-\theta t}) = \exp(-e^{-T}) \quad (n \gg 1): \quad (5.59)$$

the conditioned rupture time converges to the standard Gumbel law. Burst sizes select burst times from a travelling window of width of order $1/\theta$ around $(\log n)/\theta$, and the window does not widen as n grows.

5.7.3 The mean burst time when death is possible

When $\mu > 0$ the burst time has no finite mean, because the mass b at $\tau = \infty$ makes the integral diverge. Conditioned on bursting it is finite and explicit.

Proposition 5.13 (Mean burst time conditioned on bursting). *For $\mu > 0$,*

$$\mathbb{E}[\tau \mid \text{burst}] = \frac{1}{\lambda(1-b)} \log \frac{a-b}{a-1}. \quad (5.60)$$

Proof. The density of τ on the bursting realisations is $\varphi = \delta J$, with total mass $1-b$, so $\mathbb{E}[\tau \mid \text{burst}] = (1-b)^{-1} \int_0^\infty t \varphi(t) dt$. Since $\varphi = -(I-b)'$, integrating by parts on $[0, T]$ gives

$$\int_0^T t \varphi(t) dt = -T(I(T) - b) + \int_0^T (I(t) - b) dt,$$

and the boundary term vanishes as $T \rightarrow \infty$ because $I(T) - b = B(a-b)/(B+Aw)$, read off (5.8), decays like $e^{-\theta T}$. With $dt = dw/(\theta w)$ and $1/[w(B+Aw)] = B^{-1}(1/w - A/(B+Aw))$,

$$\int_0^\infty (I(t) - b) dt = \frac{B(a-b)}{\theta} \int_1^\infty \frac{dw}{w(B+Aw)} = \frac{1}{\lambda} \log \frac{a-b}{a-1},$$

and dividing by $1 - b$ gives (5.60). $\square$

This is not the mean productive lifetime $\mathbb{E}[T_{\text{prod}}] = \lambda^{-1} \log \frac{a}{a-1}$ of [proposition 5.8](#), and the distinction should be kept visible: $\mathbb{E}[\tau \mid \text{burst}]$ averages the rupture time over the cells that burst, while $\mathbb{E}[T_{\text{prod}}]$ averages the time spent productive over *all* cells, the internally cleared ones included with their shorter productive spans. When $\mu = 0$ the two coincide, both reducing to $\lambda^{-1} \log(1 + \lambda/\delta)$; that coincidence supports less than it appears to, and [section 5.8](#) shows what goes wrong when it is leaned on.

5.8 Conditional burst means

Three different averages of the burst size appear in this chapter. They differ by factors of nearly two and a half, and they are easy to confuse, because in ordinary usage all three are written $\mathbb{E}[\mathcal{K} \mid \cdot]$. This section separates them and quantifies the difference; the results are collected in [table 5.2](#) and plotted in [figure 5.11](#), and the chapter on population dynamics draws on all three.

5.8.1 The overall expected burst size

Whether a catastrophe has occurred by t is recorded by the truth of $W_t > 0$, the minimum burst size being 1; that event has probability $1 - I(t)$. Decomposing $\mathbb{E}[W_t]$ on it, and using $\mathbb{E}[W_t \mid W_t = 0] = 0$,

$$\mathbb{E}[W_t \mid W_t > 0] = \frac{V(t)}{1 - I(t)}. \quad (5.61)$$

This is the mean burst size given that a burst has happened by t . The expected size of an arbitrarily generated burst is its late-time value,

$$\mathbb{E}[\mathcal{K}] = \lim_{t \rightarrow \infty} \frac{V(t)}{1 - I(t)} = \frac{V_\infty}{1 - b} = \frac{\lambda - \mu}{\delta} + \frac{1}{1 - b} = \frac{a}{a - 1}, \quad (5.62)$$

the last two forms from (5.12). The distinction from V_∞ itself matters: V_∞ averages in the realisations that cleared internally and released nothing, so it is *not* the mean burst size unless $\mu = 0$.

5.8.2 The apparent circularity

What is the expected burst size of a cell that has *not yet* burst? Decomposing $\mathbb{E}[\mathcal{K}]$ on the event $\{\tau > t\}$,

$$\begin{aligned} \mathbb{E}[\mathcal{K}] &= \mathbb{E}[\mathcal{K} \mid \tau > t] \mathbb{P}\{\tau > t\} + \mathbb{E}[\mathcal{K} \mid \tau < t] \mathbb{P}\{\tau < t\} \\ &= \mathbb{E}[\mathcal{K} \mid \tau > t] I(t) + \frac{V(t)}{1 - I(t)} (1 - I(t)), \end{aligned} \quad (5.63)$$

so that

$$\mathbb{E}[\mathcal{K} \mid \tau > t] = \frac{\mathbb{E}[\mathcal{K}] - V(t)}{I(t)} = \frac{V_\infty/(1 - b) - V(t)}{I(t)}. \quad (5.64)$$

Letting $t \rightarrow \infty$ in (5.64) appears to give

$$\lim_{t \rightarrow \infty} \mathbb{E}[\mathcal{K} \mid \tau > t] = \frac{V_\infty/(1 - b) - V_\infty}{b},$$

which after a little algebra returns $\mathbb{E}[\mathcal{K}]$ itself. That cannot be right. Conditioning on $\tau > t$ as $t \rightarrow \infty$ selects the realisations that never burst at all, and their burst size is zero, not typical.

Quantity	Definition	At (1, 0.2, 0.05)
V_∞	$\mathbb{E}[\mathcal{K} \mathbf{1}\{\text{burst}\}]$; includes non-bursters at 0	13.99
$\mathbb{E}[\mathcal{K} \mid \text{burst}]$	conditioned on eventual rupture; $= a/(a-1)$	17.23
$\lim_t \mathbb{E}[\mathcal{K} \mid \tau > t, \text{burst}]$	size-biased quasi-stationary mean; $= (a+1)/(a-1)$	33.46

Table 5.2: Three means, three conditionings. The first two differ by the factor $1 - b$, the probability of eventual rupture; the second and third by the size-biasing factor $(a + 1)/a$ of [section 5.6.3](#).

The resolution is that (5.64) mixes two conditionings. Its numerator $\mathbb{E}[\mathcal{K}] - V(t)$ is the contribution to the burst size from realisations that burst *after* t ; its denominator $I(t)$ is the probability of not having burst by t , which includes the mass b of realisations that never will and whose burst size is zero. Written consistently,

$$\mathbb{E}[\mathcal{K} \mathbf{1}\{\text{burst}\} \mid \tau > t] = \frac{V_\infty - V(t)}{I(t)} \xrightarrow{t \rightarrow \infty} \frac{V_\infty - V_\infty}{b} = 0, \quad (5.65)$$

which is correct and unmysterious. The genuinely interesting limit conditions on eventual burst as well:

$$\mathbb{E}[\mathcal{K} \mid \tau > t, \text{burst eventually}] = \frac{V_\infty - V(t)}{I(t) - b}, \quad (5.66)$$

of the form $0/0$ as $t \rightarrow \infty$. By l'Hôpital's rule, with $V' = \delta K$ and $I' = -\delta J$,

$$\lim_{t \rightarrow \infty} \mathbb{E}[\mathcal{K} \mid \tau > t, \text{burst eventually}] = \lim_{t \rightarrow \infty} \frac{K(t)}{J(t)} = \frac{\langle X^2 \rangle_{\text{QS}}}{\langle X \rangle_{\text{QS}}} = 1 + \frac{2\lambda}{\delta}(1 - b), \quad (5.67)$$

in agreement with (5.52): the eventual burst size of a very patient cell is the size-biased quasi-stationary mean.

No interchange of limits is needed to reach it. The defect cancels in the ratio, so K/J is at every time the ratio of the second conditional moment to the first, and by (5.36) that conditional law is geometric with ratio $P(t)$; hence

$$\frac{K(t)}{J(t)} = \frac{1 + P(t)}{1 - P(t)}, \quad (5.68)$$

an identity holding at every t , not only in the limit. (5.67) is the value of (5.68) at the endpoint $P = 1/a$, and the only limit taken anywhere in the argument is $P(t) \uparrow 1/a$ from [theorem 5.4](#).

Remark 5.14 (Where the circularity was). The circularity was never in the mathematics; it was in the conditioning. Each of the three quantities in [table 5.2](#) is a correct answer to a different question, and only the middle one is what is usually meant by “the mean burst size”. An assay that observes a cell because it ruptured measures the second; an assay that observes cells still infected at a late time and waits measures the third.

Three means, and with them the cell founded by one particle is completely described. The two sections that follow ask how much of that survives when the founder is not alone, and when the cell is not.

5.9 Several founders

The single-cell law of [sections 5.4](#) and [5.6](#) was derived from one founder. The first probe of how far it reaches asks what happens when there are several.

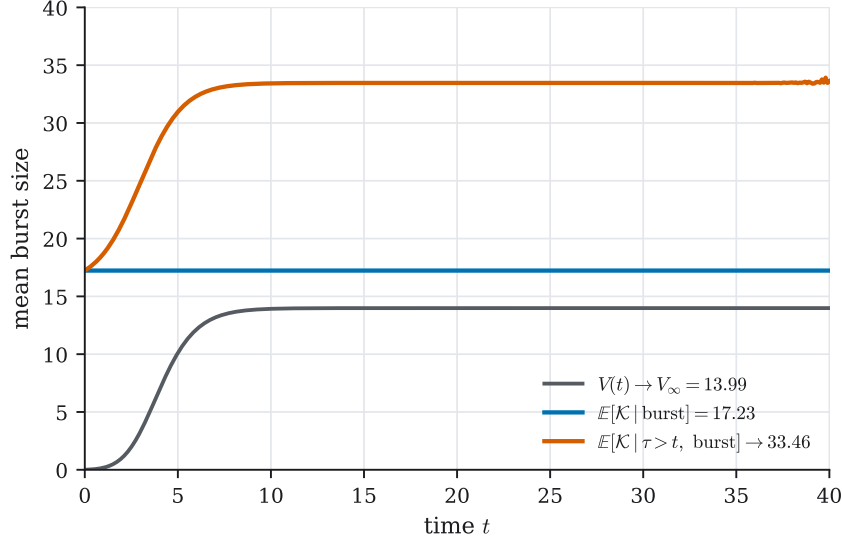


Figure 5.11: Three distinct averages at $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, drawn as trajectories. The unconditional mean release $V(t)$ approaches $V_\infty = 13.99$ and includes cells that never burst; the burst mean $V_\infty/(1-b) = a/(a-1) = 17.23$ is the horizontal line; conditioning on a very late eventual burst approaches the size-biased limit $(a+1)/(a-1) = 33.46$. Keeping the conditioning explicit removes the apparent circularity.

In applications a single founder is not always the right picture: a macrophage may be invaded by several bacteria at once, and in the laboratory the multiplicity of infection is a control parameter. Nothing essential changes — the process is still linear and $G_k = G^k$ still holds — but the bookkeeping does, and two plausible guesses turn out to be wrong. Throughout, k founders share one cell and hence one catastrophe clock: rupture occurs at rate δ times the *total* load, and dumps the whole cell.

5.9.1 Fixation functions for k founders

By the branching property the k founder lineages evolve independently until the shared clock rings. No burst occurs by t precisely when no lineage has burst; internal extinction occurs precisely when every lineage has cleared internally. Hence

$$I_k(t) = I(t)^k, \quad D_k(t) = D(t)^k, \quad I_{\text{fix},k}(t) = I(t)^k - D(t)^k. \quad (5.69)$$

Remark 5.15 (A guess that fails). The tempting guess is $I_{\text{fix},k} = I_{\text{fix}}^k$: each lineage stays productive with probability I_{fix} , independently. But “productive” for the whole cell means *not fixed*, and the intersection of the independent events “lineage i avoids bursting and does not go extinct” is not the event “the cell is not fixed”: the cell also stops being productive when every lineage dies out. Correctly, $I_{\text{fix},k} = I^k - D^k$, whereas $I_{\text{fix}}^k = (I - D)^k$; already for $k = 2$ the two differ by $I^2 - D^2 - (I - D)^2 = 2D(I - D) \neq 0$ whenever $\mu > 0$. Numerically, at $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ and $t = 1$: the correct $I_{\text{fix},2} = I^2 - D^2 \approx 0.8458$, while $I_{\text{fix}}^2 \approx 0.6542$. When $\mu = 0$, $D \equiv 0$ and the two coincide — again the invisible case.

5.9.2 Moments and the release kernel

The moments fall out of the k -founder PGF of [section 5.3.3](#), $G_k = G^k$, where G is the single-founder defective PGF. Differentiating at $z = 1$, with $G(1, t) = I(t)$, $\partial_z G|_{z=1} = J(t)$ and $\partial_z^2 G|_{z=1} = K(t) - J(t)$,

$$J_k(t) = k I(t)^{k-1} J(t), \quad (5.70)$$

and, from the second factorial moment,

$$\begin{aligned}
K_k(t) &= \partial_z^2 G_k|_{z=1} + \partial_z G_k|_{z=1} \\
&= k(k-1)I(t)^{k-2}J(t)^2 + kI(t)^{k-1}(K(t) - J(t)) + kI(t)^{k-1}J(t) \\
&= kK(t)I(t)^{k-1} + k(k-1)J(t)^2I(t)^{k-2}.
\end{aligned} \tag{5.71}$$

Multiplying by the catastrophe rate turns K_k into the expected release flux — the second item of the consignment of [section 5.1.3](#), and the kernel on which the renewal model of the chapter on population dynamics is built:

$$g_k(t) = \delta K_k(t) = \delta \left(kK(t)I(t)^{k-1} + k(k-1)J(t)^2I(t)^{k-2} \right). \tag{5.72}$$

[Figure 5.13\(a\)](#) plots g_k against age for $k = 1, \dots, 4$. The ordering in k is superlinear early and reverses late, so the enhancement is a property of the cell's age rather than a fixed multiplier.

Remark 5.16 (The free-sum trap). It is tempting to write $g_k = \delta (kK + k(k-1)J^2)$, the second moment of a free sum of k independent copies. That form omits the powers of I : the second-moment contribution of a configuration is counted only while the shared clock has *not* rung. At $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$, $t = 1$, $k = 2$: the free-sum form gives $K_2 \approx 23.39$, while (5.71) gives $K_2 \approx 22.264$, which agrees with a direct master-equation computation. The difference is the superlinearity of g_k in k wearing a disguise: even the free-sum form is superlinear, but the true kernel is superlinear *and* tempered by survival. That a cell's expected release flux depends superlinearly on its founding multiplicity is a genuine prediction of the model — one that classical constant- p models cannot express, and that low-multiplicity single-cell assays can test.

5.9.3 Lifetime yield

The lifetime mean release from k founders is $V_\infty^{(k)} = \int_0^\infty g_k(t) dt$. Carrying out the integration, which is deferred to [section 5.C.1](#),

$$V_\infty^{(1)} = V_\infty = \frac{a(1-b)}{a-1}, \tag{5.73}$$

and, for $k \geq 2$,

$$V_\infty^{(k)} = (1-b^k) + \frac{2\lambda}{\delta} \left[\frac{1}{k+1} - b^k + \frac{k b^{k+1}}{k+1} \right] + k(k-1) \frac{\lambda}{\delta} \int_1^b (x-a)(x-b) x^{k-2} dx. \tag{5.74}$$

In the no-death case the answer collapses:

$$V_\infty^{(k)} = k + \frac{\lambda}{\delta} \quad (\mu = 0). \tag{5.75}$$

Indeed, when $\mu = 0$ the event type is independent of the load ([section 5.10.1](#)), so the number of births before the single catastrophe is $\text{Geom}_0(s)$ with mean $\rho/s = \lambda/\delta$ whatever the founding load, and the total release is simply k founders plus that many births. With $\lambda = 1$ and $\delta = 0.1$, for instance, $V_\infty^{(k)} = 11, 12, 13, 14, 15, \dots$

Two consequences are worth pausing on, and they pull in opposite directions. The yield is *not* kV_∞ : the first catastrophe dumps the entire cell once, so additional founders add their own expected descendants but not another cell's worth of release. At $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ a naive scaling would predict $2V_\infty \approx 27.97$ for two founders; the true value is $V_\infty^{(2)} \approx 17.43$, and [figure 5.12](#) shows the gap widening in k . Yet the *conditional* mean burst size given eventual burst, $V_\infty^{(k)}/(1-b^k)$, *increases* with k — 17.23, 18.07, 19.02, 21.00 for $k = 1, 2, 3, 5$ at the same

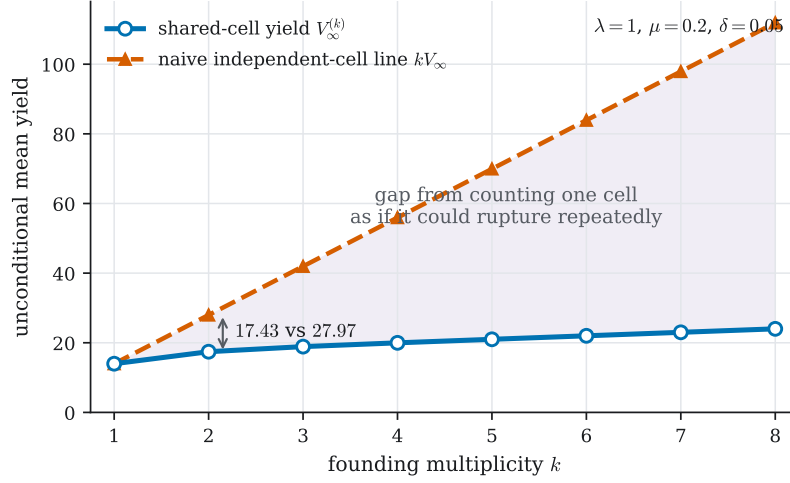


Figure 5.12: Unconditional mean yield $V_\infty^{(k)}$ against founding multiplicity k , compared with the naive independent-cell line kV_∞ , for $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$. The true yield is sub-extensive because all founders share one cell and hence one catastrophe; for $k = 2$, $V_\infty^{(2)} = 17.43$ rather than $2V_\infty = 27.97$.

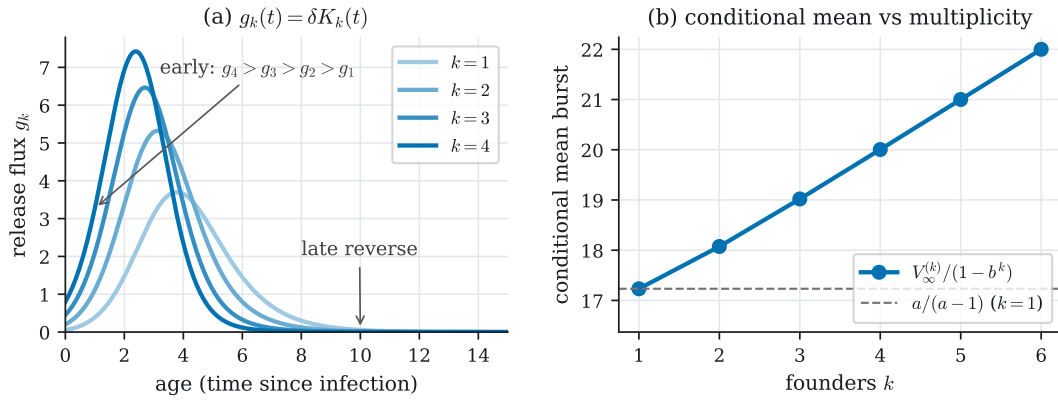


Figure 5.13: Multiplicity of infection at $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$. (a) Release fluxes $g_k(t) = \delta K_k(t)$ from (5.72) for $k = 1, \dots, 4$: early ages are ordered superlinearly in founding load, while late fluxes reverse as multi-founder lineages fix faster, so the enhancement is age-dependent rather than a pointwise ordering valid for all t . (b) Conditional mean burst $V_\infty^{(k)}/(1 - b^k)$ increases with k ; the dashed line is the $k = 1$ value $a/(a - 1) = 17.23$.

parameters. Multiply infected cells are more likely to burst at all, and when they do, they burst bigger. The two statements are consistent because the first is unconditional and the second is not, which is table 5.2's distinction appearing again in a new place.

The geometric structure of theorem 5.4 survives all of this untouched, the load in a multiply infected cell being a sum of independent copies of the single-founder load. Multiplicity changes the bookkeeping, not the shape; section 5.10 finds the boundary the other way.

5.10 Chained immediate transfer

The second probe couples cells together in the most austere way available. Take M host cells, all empty but one, which holds a single particle. Each runs the $\mu = 0$ birth–catastrophe process of the previous sections, and on rupture the released load does not enter an extracellular pool:

it passes immediately and wholesale into a fresh cell, which begins the process again with that load as its founding census, and so on until every cell has ruptured. There is no extracellular phase, no clearance and no re-infection, only a chain of bursts; and since $\mu = 0$, all M ruptures occur almost surely.

Serial rupture of this kind is not a fiction: the chapter on bursting and budding pathogenesis gives the biological account, with its Brownian-drift and macrophage-motility caveats. Here it is a tractable laboratory for multi-cell burst statistics, interpolating between the single-cell theory above and the populations of the chapter on population dynamics.

Let r_i be the load at the i th rupture, t_i its time, and $T_i = t_i - t_{i-1}$ the i th inter-rupture interval, with $t_0 = 0$. Write

$$s = \frac{\delta}{\lambda + \delta}, \quad \rho = \frac{\lambda}{\lambda + \delta} = 1 - s. \quad (5.76)$$

5.10.1 The key observation: event types ignore the load

When the load is n , the next event is a birth with probability $\lambda n / (\lambda n + \delta n) = \rho$ and a catastrophe with probability s , independent of n ; only the holding time depends on the load, being $\text{Exp}((\lambda + \delta)n)$. The sequence of event *types* is therefore independent coin flips, and the number of births G before the catastrophe in a cell founded with any load is

$$G \sim \text{Geom}_0(s), \quad \mathbb{P}\{G = g\} = s \rho^g, \quad g = 0, 1, 2, \dots, \quad (5.77)$$

the number of failures before the first success. Each cell in the chain contributes an *independent* copy G_i , since the process restarts afresh at every transfer, and the rupture size of cell i is its founding load plus its own births. The first cell being founded by a single particle,

$$r_k = 1 + G_1 + \dots + G_k, \quad (5.78)$$

with $G_1, \dots, G_k$ independent and identically distributed. Everything about the rupture sizes follows from this decomposition, and something has quietly happened in reaching it: the load, the whole subject of [section 5.4](#), has dropped out of the argument.

5.10.2 Rupture-size laws

A sum of k independent geometric variables is negative binomial, so (5.78) gives immediately

$$\mathbb{P}\{r_k = n\} = \mathcal{A}_k^{(n)} s^k \rho^{n-1}, \quad \mathcal{A}_k^{(n)} = \binom{n+k-2}{k-1}, \quad n \geq 1. \quad (5.79)$$

The first few cases show the pattern:

$$\mathbb{P}\{r_1 = n\} = s \rho^{n-1}, \quad (5.80)$$

geometric, matching the one-cell law of (5.46) at $\mu = 0$, and then

$$\mathbb{P}\{r_2 = n\} = n s^2 \rho^{n-1}, \quad \mathbb{P}\{r_3 = n\} = \frac{n(n+1)}{2} s^3 \rho^{n-1}, \quad \mathbb{P}\{r_4 = n\} = \frac{n(n+1)(n+2)}{6} s^4 \rho^{n-1}. \quad (5.81)$$

Equivalently the coefficients satisfy

$$\mathcal{A}_k^{(n)} = \sum_{i=1}^n \mathcal{A}_{k-1}^{(i)}, \quad \mathcal{A}_1^{(i)} = 1, \quad (5.82)$$

which is (5.79) conditioned on the value of r_{k-1} : $\mathbb{P}\{r_k = n\} = \sum_i \mathbb{P}\{r_{k-1} = i\} \mathbb{P}\{G_k = n - i\}$. The closed form then follows by induction from the hockey-stick identity.

Remark 5.17 (The coefficient). The coefficient in (5.79) is easy to miscount. The superficially similar expression $\frac{1}{k!} \frac{(n+k-1)!}{(n-1)!} = \binom{n+k-1}{k}$ is off by one shift, and is already contradicted by the r_2 and r_3 cases displayed above, which (5.79) reproduces exactly. The correct coefficient is $\binom{n+k-2}{k-1}$, and it has a direct reading: it counts the ways of distributing the $n-1$ births accumulated along the chain among its k cells, each cell receiving any non-negative number — the stars-and-bars count for (5.78).

Further consequences, all verified numerically in [section 5.B](#):

$$\mathbb{E}[r_k] = 1 + k \frac{\rho}{s} = 1 + k \frac{\lambda}{\delta}, \quad \text{Var}(r_k) = k \frac{\rho}{s^2}, \quad (5.83)$$

$$\mathbb{E}[z^{r_k}] = z \left(\frac{s}{1 - \rho z} \right)^k, \quad \sum_{n \geq 1} \mathbb{P}\{r_k = n\} = 1. \quad (5.84)$$

The mean rupture size grows linearly along the chain: each cell adds one expected burst's worth of births, λ/δ , to what it inherited. [Figure 5.14](#) shows the laws against exact simulation, with the inter-rupture intervals, which shorten as the inherited load grows.

5.10.3 Rupture times

The first rupture time inherits the single-cell flux. With m founders the no-rupture probability is $I(t)^m$ by (5.69), since $D \equiv 0$ at $\mu = 0$, so

$$\mathbb{P}\{t_1 > t\} = I(t)^m, \quad f_{t_1}^{(m)}(t) = -m I(t)^{m-1} I'(t) = m I(t)^{m-1} \delta J(t). \quad (5.85)$$

For $m = 1$ this is the burst-time density $\varphi = \delta J$ of (5.53), proper here because $b = 0$. When $\mu > 0$ the same reasoning must distinguish the two fixation routes: the *burst-only* time has distribution function $(1 - I(t))/(1 - b)$ and density $\delta J(t)/(1 - b)$, while the *general-fixation* time — rupture or internal extinction — has density

$$\delta J(t) + \mu p_1(t), \quad (5.86)$$

the extra term being the flux through $1 \rightarrow 0$: only a population of size exactly one can die out in a single death event.

The intervals, their Laplace transforms and the joint structure of the second rupture time are collected in [section 5.C.3](#), beside the hypergeometric machinery they need.

5.10.4 Where the structure breaks

That extra term in (5.86) is the whole difficulty. The $\mu = 0$ chain rests on (5.77), and the event-type decomposition works precisely because a cell then has one way to stop: it ruptures. Once $\mu > 0$ a lineage can vanish internally, event types stop being independent of the load, and (5.78) has no analogue. The joint laws of (t_k, r_k) for general $\mu > 0$ therefore remain open; [section 5.C.2](#) sets out the available route and [section 5.12.4](#) states what is missing. For $\mu = 0$, by contrast, (5.78) extends the size laws to all M cells at once, nothing in it depending on where a cell sits in the chain.

This contrast is why both probes were run. $G_k = G^k$ needs only independence between lineages, and multiplicity supplies it; the chain's decomposition needs the stopping mechanism to be unique, and internal extinction destroys that.

5.11 Bursting against budding

This chapter opened by objecting to the constant drip. It can now say precisely what that objection amounts to, and at the level where it was made: one cell.

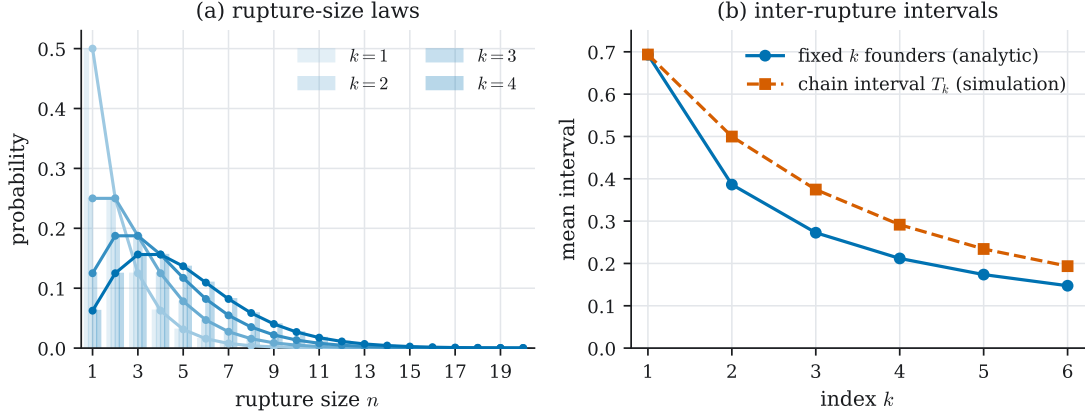


Figure 5.14: Chained immediate transfer with $(\lambda, \delta) = (1, 1)$ and $s = \rho = 1/2$. (a) Simulated rupture-size probabilities (bars) from 10^6 chain realisations with seed 42, with the exact negative-binomial laws (5.79) (curves) for $k = 1, \dots, 4$. (b) The analytic mean interval $\mathbb{E}[T(k)]$ of (5.96) at fixed founding load k , and the simulated mean interval at chain position k ; both decrease over $k = 1, \dots, 6$, the latter faster because its founding loads are random and growing.

The comparator is the budding cell of the constant-rate models. In its simplest stochastic form a budding infected cell experiences two *independent* event types — release of one infectious particle at rate p , and death of the cell at rate $d_{\mathcal{I}}$ — and the two continue until the cell dies. Its survival and mean release are

$$I_{\text{bud}}(t) = e^{-d_{\mathcal{I}}t}, \quad V_{\text{bud}}(t) = \frac{p}{d_{\mathcal{I}}} \left(1 - e^{-d_{\mathcal{I}}t}\right), \quad (5.87)$$

the same one-cell limit the Basic Model of Viral Replication reaches at the population level. The release count of a budding cell is geometric on $\{0, 1, 2, \dots\}$:

$$\mathbb{P}\{\mathcal{K}_{\text{bud}} = k\} = \frac{d_{\mathcal{I}}}{d_{\mathcal{I}} + p} \left(\frac{p}{d_{\mathcal{I}} + p}\right)^k, \quad k = 0, 1, 2, \dots, \quad (5.88)$$

since each successive event is a release with probability $p/(d_{\mathcal{I}} + p)$ and the terminating death with probability $d_{\mathcal{I}}/(d_{\mathcal{I}} + p)$.

The structural contrast with bursting is exact: in budding, release and death are independent events, and a cell may release many particles and then die with nothing left to release; in bursting, release and death are the *same* event, and whatever has accumulated is released at once. That difference is why a bursting cell's release law is a statement about its accumulated load, and why section 5.6 has no budding analogue: a cell that releases continuously holds nothing to be quasi-stationary about. Table 5.3 sets the two descriptions side by side for the $\mu = 0$ bursting case.

Two rows of the table carry the comparison. The mean cell lifetime of a bursting cell grows only logarithmically with λ/δ while its mean release grows linearly; budding offers no such coupling, its lifetime and its release being controlled by separate parameters. And the release laws, both geometric, differ in their support: budding can release nothing at all from a cell that died early, while bursting never releases nothing from a cell that ruptured. A trade-off along an axis of this kind is a familiar theme in the evolution of virulence [9].

Matching the means does not repair the difference. Figure 5.15 sets the two release laws against each other with the budding mean tuned to the BDC conditional mean $V_{\infty}/(1 - b) = 17.23$: the budding law still holds probability 0.055 of releasing nothing. Matching one moment matches neither the support nor the tails. That is the concrete form of the objection this chapter opened

	budding	bursting ($\mu = 0$)
$\frac{dI}{dt}$	$-d_{\mathcal{I}}I$	$-(\lambda + \delta)I + \lambda I^2$
$I(t)$	$e^{-d_{\mathcal{I}}t}$	$\frac{\lambda + \delta}{\lambda + \delta e^{(\lambda + \delta)t}}$
mean cell lifetime	$1/d_{\mathcal{I}}$	$\frac{1}{\lambda} \log \left(1 + \frac{\lambda}{\delta} \right)$
$V(t)$	$\frac{p}{d_{\mathcal{I}}}(1 - I)$	$\frac{\lambda}{\delta} + 1 - \frac{2\lambda + \delta}{\delta}I + \frac{\lambda}{\delta}I^2$
release law	geometric on $\{0, 1, \dots\}$, (5.88)	geometric on $\{1, 2, \dots\}$, $\mathbb{P}\{\mathcal{K} = k\} = \frac{\delta}{\lambda}a^{-k}$

Table 5.3: Single-cell budding and bursting compared, $\mu = 0$ for bursting. Budding parameters: release rate p and infected-cell death rate $d_{\mathcal{I}}$. Bursting parameters: division rate λ and rupture rate δ per intracellular particle; $a = (\lambda + \delta)/\lambda$ here.

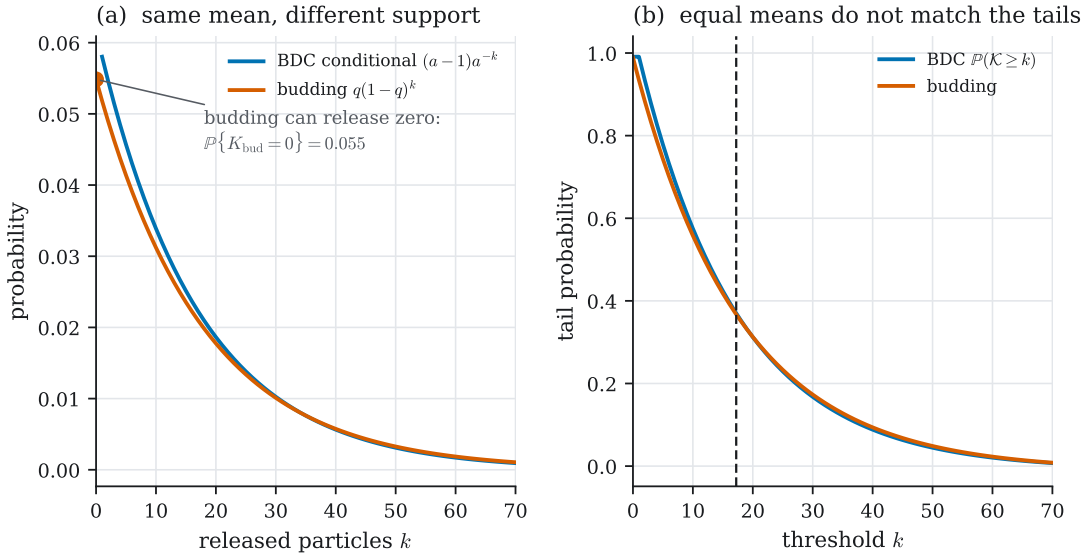


Figure 5.15: Release-count laws for the BDC process at $(\lambda, \mu, \delta) = (1, 0.2, 0.05)$ and a budding comparator whose mean $p/d_{\mathcal{I}}$ is matched to the BDC conditional mean $V_{\infty}/(1 - b) = 17.23$. Panel (a): the BDC law conditioned on bursting is geometric on $\{1, 2, \dots\}$, whereas the budding law is geometric on $\{0, 1, 2, \dots\}$ and retains probability 0.055 of zero release. Panel (b): survival functions $\mathbb{P}\{\mathcal{K} \geq k\}$ and $\mathbb{P}\{\mathcal{K}_{\text{bud}} \geq k\}$. Matching the means matches neither the support nor the tails.

with: a fitted constant-rate model can reproduce the average yield of a lytic cell and still be wrong about the quantity that decides whether a small inoculum establishes at all.

The comparison is deliberately confined to one cell. What happens when these descriptions are embedded in a population, where the timing of release governs establishment and invasion speed, belongs to the chapter on population dynamics; the pathogenesis-level comparison belongs to the chapter on bursting and budding pathogenesis. All three chapters return to this contrast, and answer different questions with it.

5.12 Discussion

The single-cell description is now complete in distribution: geometric states at every time, a geometric quasi-stationary limit, and a burst-size law identical to it, with the conditional laws of

burst time and burst size, the multi-founder extension and the chained-transfer model alongside. What follows reads that description into measurement, into two published parameter sets, and into what is still open; the population consequences are taken up in the chapter on population dynamics.

5.12.1 Single-cell assay predictions for lytic systems

For a genuinely lytic system — *Y. pestis*, *F. tularensis*, lytic phage — a limiting-dilution assay, one infected cell per well with counts followed in time, measures the single-cell process directly, with no population model interposed. The number of new infectious particles released per cell is an integer-valued random variable, and its full distribution, not merely its mean, determines whether a small inoculum grows into a detectable infection [10, 11]. The theory makes five predictions, all parameter-free given (λ, μ, δ) :

1. a fraction b of wells show no release ever — the mass of internal extinction;
2. the cumulative release in a well is a step function: zero, then a single terminal jump;
3. the jump times follow the density $\delta J(t)/(1 - b)$ of (5.53);
4. the jump sizes follow the geometric law with ratio $1/a$ — the quasi-stationary distribution, by theorem 5.10;
5. the between-well variance of total release is $\text{Var}(W_\infty)$ from (5.13), a quantity no deterministic constant-rate model can produce at all.

The five are not all of a kind. The second tests a standing assumption rather than a derived result — instantaneous complete release is assumption 3 of section 5.1.1, and an assay that saw graded release would refute the premise before reaching the theory — and the first is a one-parameter consequence of the roots alone. Predictions 3 to 5 are where the chapter’s own content is exposed: a density, a distribution and a variance, each fixed by (λ, μ, δ) with nothing left to tune. Stated as literally as that, the predictions are too strong for a budding virus: intermittent budding is not a single terminal burst [12]. They belong to lytic hosts, or to the bursting endpoint of the spectrum of release models the chapter on population dynamics lays out.

5.12.2 The same theory at two extremes

The theory has already been fitted to two intracellular pathogens whose parameter estimates sit at opposite ends of what the model allows, and the contrast shows how much behaviour is fixed by (λ, μ, δ) alone. Table 5.4 evaluates the closed forms at both.

For *F. tularensis* strain SCHU S4, modelling of murine challenge data gave $\lambda = 0.15$, $\mu = 0.01$ and $\delta = 1.5 \times 10^{-4}$ per hour [13]. At those rates fewer than 7% of infected cells fail to burst in the model, and those that rupture release a geometric number of bacteria with mean 934 and median 648. Bursts of that size sit comfortably with the observed ability of doses as low as ten colony-forming units to establish infection [15]: on the model’s account each rupture floods the local environment heavily enough that a small inoculum needs only a few rounds of the cycle to reach large numbers.

For *Bacillus anthracis*, approximate Bayesian computation against published time series produced very different estimates: $\lambda \approx 0.64$, $\mu \approx 1.64$, $\delta \approx 0.04$ per hour [14]. Here death dominates birth, and the model has more than 96% of phagocytes clearing their infection without rupturing, the rest releasing about 1.6 bacteria on average. The reported infectious dose, 8×10^3 to 5×10^4 spores [14], is orders of magnitude above the tularaemia figure, and matches a pathogen whose intracellular dynamics destroy far more than they release.

	<i>F. tularensis</i> SCHU S4	<i>B. anthracis</i>
$(\lambda, \mu, \delta), \text{h}^{-1}$	$(0.15, 0.01, 1.5 \times 10^{-4})$	$(0.64, 1.64, 0.04)$
a	1.001	2.66
b , never bursts	6.66 %	96.24 %
$\mathbb{E}[\mathcal{K} \mid \text{burst}] = a/(a - 1)$	934	1.60
median burst size	648	1
$\mathbb{E}[T_{\text{prod}}]$	45.6 h	0.74 h
$\mathbb{E}[\tau \mid \text{burst}]$	48.4 h	0.93 h

Table 5.4: The same three-rate theory at two published parameter sets, from [13] and [14]. Every entry is a closed form of this chapter evaluated at those rates: (5.2), (5.12), (5.35), (5.42) and (5.60). The digits shown are those of the evaluation, not of the estimate: each entry is exact given the rates in the first row, and inherits whatever uncertainty those rates carry. The two columns come from different methods applied to different data, so the comparison between them is qualitative even where the arithmetic within a column is not.

Two rows of table 5.4 are directly falsifiable, and they are the rows the chapter had not previously computed. On these estimates a *F. tularensis*-infected macrophage should rupture at around two days post-infection, and an anthrax-infected phagocyte, in the rare event that it ruptures at all, within the hour. Neither number has a measurement to sit beside it here. What would settle them is time-lapse imaging of infected macrophages held to rupture, with the time of the release event recorded and the payload counted directly; the prediction is sharp enough that a handful of such recordings would be decisive, and sharp enough to be wrong. Both timings are only as good as the rates they condition on, and those rates came from different methods and different data. What the pair shows regardless is one three-rate theory spanning the space from near-certain large bursts to near-certain clearance with minimal release.

5.12.3 What the two probes showed

The two probes of sections 5.9 and 5.10 were asked the same question — how far does the geometric structure reach — and returned opposite answers, which is the more informative outcome.

Multiplicity leaves the structure intact. Lineages are independent until the shared clock rings, so $G_k = G^k$ and the load in a multiply infected cell is a sum of independent copies of the single-founder load; the bookkeeping changes, and two natural guesses about it fail (remarks 5.15 and 5.16), but the geometric shape does not. Chaining destroys it as soon as internal extinction is possible, because the $\mu = 0$ decomposition needs a cell to have exactly one way to stop and the death rate gives it two. *The geometric structure is robust to multiplicity and fragile to internal extinction in the chained setting*, and the two failures are failures of different things — a counting identity in one case, a decomposition of the process in the other.

5.12.4 Open problems

1. **How far the telescoping criterion reaches.** Theorem 5.10 closes what was an open problem: the identity between burst size and the quasi-stationary law has a structural proof, and remark 5.11 states the property that proof uses. Two questions survive it. Does that property *characterise* the processes for which the identity holds, or is it merely sufficient? And can the identity be obtained independently by a Yaglom-type argument [16], which would connect it to the existence theory rather than to the calculus?
2. **Chained joint laws for general μ .** The joint laws of (t_k, r_k) for $\mu > 0$, and the joint law

of (t_M, r_M) for the full M -cell chain, remain open. The coefficient machinery of [section 5.C.2](#) is the available route, and [section 5.10.4](#) says why the elementary route is not.

3. **Logistic intracellular growth.** The standing objection to linear birth is that it lets the conditional load grow without bound. The defence is that the burst hazard is proportional to load and so regulates it — [theorem 5.6](#) is precisely the statement that the conditional load stabilises at $a/(a-1)$ — but the sensitivity of the burst statistics, and of the kernels the chapter on population dynamics will use, to a logistic birth rate remains to be checked numerically.

5.12.5 What comes next

Everything in this chapter is single-cell. The released particles have so far been counted and then forgotten, and that is the approximation the chapter on population dynamics removes. There the survival and release statistics derived here become the age-structured kernels of a population-level renewal model: cells infected at different times contribute to the infected-cell and free-particle pools according to $I_{\text{fix}}(a)$ and $g(a) = \delta K(a)$, in the tradition of age-structured renewal running from [\[17, 18\]](#). The constant-rate models of viral dynamics then appear as a projection of that renewal system, and bursting is compared against budding at the level where the comparison matters — establishment, invasion speed, and what data can and cannot identify.

One notational matter is left for later. The two-type chapter solves a Riccati equation structurally identical to [\(5.7\)](#) under unrelated names, as [section 5.2.3](#) noted; aligning the two would mean renaming a and b throughout this chapter, the chapter on the birth–death–catastrophe process and the chapter on population dynamics, and is a thesis-wide decision rather than a local one.

5.A Master formula table: single cell

Throughout: $w = e^{\theta t}$, $\theta = \lambda(a-b)$, $A = a-1$, $B = 1-b$, $\kappa = 1 + \delta/(2\lambda)$, $s = \delta/(\lambda + \delta)$, $\rho = 1-s$. The provenance column reads **R** for a result recalled from the chapter on the birth–death–catastrophe process, **M** for one taken from the mathematical introduction, **S** for one standard in the literature, and **N** for one established in this chapter. [Table 5.5](#) collects them.

Table 5.5: Every closed form the chapter uses, with its provenance.

Quantity	Closed form	Prov.
Roots a, b	$\eta \pm \sqrt{\eta^2 - \mu/\lambda}$, $\eta = (\lambda + \mu + \delta)/(2\lambda)$, $b < 1 < a$	R
A, B, θ	$a-1, 1-b, \lambda(a-b)$; $AB = \delta/\lambda$, $A+B = a-b$	R
$I(t)$	$(aB + bAw)/(B + Aw)$	R
$D(t)$	$ab(w-1)/(aw-b)$	R
$I_{\text{fix}}(t)$	$(a-b)^2 w / [(B + Aw)(aw-b)]$	R
$J(t)$	$(a-b)^2 w / (B + Aw)^2 = -\frac{\lambda}{\delta}(I-a)(I-b)$	R
$K(t)$	$[1 + 2\lambda(1-I)/\delta]J = \frac{2\lambda}{\delta}(\kappa - I)J$	R
$V(t)$	$(1-I)[1 + \lambda(1-I)/\delta]$	R
V_∞	$a(1-b)/(a-1) = aB/A$	R
$\mathbb{E}[\mathcal{K} \mid \text{burst}]$	$a/(a-1)$	R
$\mathbb{E}[W_t^2]$	$\frac{2(\lambda-\mu)}{\delta}V - \frac{\lambda+\mu}{\delta}I - K + \frac{\lambda+\mu}{\delta} + 1$	R
$G(z, t)$	$[ab(1-w) - z(a-bw)] / [(b-aw) - z(1-w)]$	S
$G_k(z, t)$	$G(z, t)^k$	S
$\varphi(t)$ (burst time)	$\delta J(t)$, total mass $1-b$	S
budding $\mathbb{P}\{\mathcal{K}_{\text{bud}} = k\}$	$\frac{d_{\mathcal{I}}}{d_{\mathcal{I}}+p} \left(\frac{p}{d_{\mathcal{I}}+p} \right)^k$, $k \geq 0$	S
$G_{\text{kill}} - G_{\text{cat}}$	$1 - I(t)$, exactly, for all z and t	N

continued overleaf

Table 5.5, continued

Quantity	Closed form	Prov.
$P(t)$	$(1-w)/(b-aw)$; $P(0) = 0$, $P(\infty) = 1/a$	N
$p_n(t)$, $n \geq 1$	$p_1(t)P(t)^{n-1}$, $p_1 = \left(\frac{a-b}{b-aw}\right)^2 w = P'/\lambda$	N
QSD(n)	$(a-1)a^{-n}$, all $\mu \geq 0$	N
ϑ (decay rate)	$\mu\nu_1 + \sum_i \delta_i \nu_i = \lambda(a-b) = \theta$	N
$\mathbb{P}\{\mathcal{K} = k\}$	$(\delta/\lambda)a^{-k}$, $k \geq 1$	N
$\mathbb{P}\{\mathcal{K} = k \mid \text{burst}\}$	$(a-1)a^{-k}$ — the QSD	N
$\mathbb{E}[\mathcal{K} \mid \tau = t]$	$K/J = 1 + \frac{2\lambda}{\delta}(1-I) = \frac{1+P}{1-P} \rightarrow (a+1)/(a-1)$	N
$\mathbb{E}[T_{\text{prod}}]$	$\lambda^{-1} \log(a/(a-1)) = \lambda^{-1} \log\langle X \rangle_{\text{QS}}$	N
$\mathbb{E}[\tau \mid \mathcal{K} = n]$ ($\mu = 0$)	H_n/θ , $\theta = \lambda + \delta$	N
$\mathbb{P}\{\tau < t \mid \mathcal{K} = n\}$ ($\mu = 0$)	$(1 - e^{-\theta t})^n \rightarrow \text{Gumbel}$	N
$\mathbb{E}[\tau \mid \text{burst}]$ ($\mu > 0$)	$\frac{1}{\lambda(1-b)} \log \frac{a-b}{a-1}$	N
$I_{\text{fix},k}(t)$	$I(t)^k - D(t)^k$	N
$J_k(t)$, $K_k(t)$	kJI^{k-1} ; $kKI^{k-1} + k(k-1)J^2I^{k-2}$	N
$g_k(t)$	$\delta K_k(t)$	N
$V_\infty^{(k)}$, $\mu = 0$	$k + \lambda/\delta$	N
r_k law ($\mu = 0$, chain)	$\binom{n+k-2}{k-1} s^k \rho^{n-1}$	N
$\mathbb{E}[r_k]$, $\text{Var}(r_k)$	$1 + k\lambda/\delta$; $k\rho/s^2$	N
$\tilde{T}(k, u)$ (interval)	$\frac{s^k}{k+u'} {}_2F_1(k+1, 1; k+1+u'; \rho)$, $u' = u/(\lambda + \delta)$	N

The $\mu = 0$ specialisation. $b = 0$, $a = 1 + \delta/\lambda$, $D \equiv 0$, $I_{\text{fix}} = I$, $V_\infty = 1 + \lambda/\delta = \langle X \rangle_{\text{QS}}$, $\mathbb{E}[T_{\text{prod}}] = \lambda^{-1} \log(1 + \lambda/\delta)$, $V_\infty^{(k)} = k + \lambda/\delta$, and $\mathbb{P}\{\mathcal{K} = k\} = s\rho^{k-1}$ with $s = \delta/(\lambda + \delta)$, $\rho = 1 - s$: the burst size is geometric on $\{1, 2, \dots\}$ with success probability s , and every realisation bursts. Several coincidences hold only here — $V_\infty = \mathbb{E}[\mathcal{K} \mid \text{burst}]$, and $\mathbb{E}[\tau \mid \text{burst}] = \mathbb{E}[T_{\text{prod}}]$ — and none of them survives $\mu > 0$.

5.B Verification record: chained immediate transfer

Every quantitative claim of [section 5.10](#) and [section 5.C.3](#) was checked by exact simulation of the chained process. Sampling is direct, because the event-type decomposition of [section 5.10.1](#) makes the sequence of event types a sequence of independent coin flips and the holding times exponential of known rate. This appendix records the suite, its tally, and how to reproduce it.

The suite `verify_chained_transfer.py` samples the chained process exactly and checks every quantitative claim across three parameter sets, $(\lambda, \delta) \in \{(1, 1), (1, 0.1), (2, 0.5)\}$: 28 checks, all passing.

ID	Name	What is checked	Result
A	Coefficient identity	$\binom{n+k-2}{k-1}$ obeys the recurrence (exact integers)	1/1
B	Rupture-size laws	$\mathbb{P}\{r_k = n\}$ vs 10^6 chains, $k = 1..4$, SE-unit comparison	3/3
C	Moments	$\mathbb{E}[r_k] = 1 + k\lambda/\delta$, $\text{Var}(r_k) = k\rho/s^2$	3/3
D	PGF	$\mathbb{E}[z^{r_k}] = z(s/(1-\rho z))^k$ (SE units)	3/3
E	First rupture time	histogram vs $\delta J(t)$	3/3
F	Interval transform	fixed founders $k \in \{1, 2, 3, 5\}$: empirical Laplace transform vs the series and its ${}_2F_1$ form	3/3
F2	Interval mixture	chain intervals T_2, T_3 (random founders) vs pmf-weighted $\tilde{T}$	3/3
G	Second rupture time	empirical $\mathbb{E}[e^{-ut_2}]$ vs $\tilde{t}_2(u)$	3/3
H/H2	t_2 density and mean	exact hypoexponential-mixture density vs histogram (relative $L^2 \approx 3\%$); exact $\mathbb{E}[t_2]$	3/3
I	Interval means	strictly decreasing; simulation vs the exact series	3/3
Total			28/28

The ${}_2F_1$ cross-check in test F activates only when `scipy` is available; the series itself, which the simulation is compared against, needs only `numpy`.

Reproduction. The suite, its report, its metrics and its figures are shipped with this chapter's source.

```
cd verification
python3 verify_chained_transfer.py
# writes verify_chained_transfer_report.txt, metrics, figures
```

What is not covered. The suite is confined to $\mu = 0$, where the event-type decomposition holds. The joint laws of (t_k, r_k) for general $\mu > 0$, and the joint law of (t_M, r_M) for the full M -cell chain, are neither derived nor checked here; see [section 5.12.4](#). For $\mu = 0$ the decomposition (5.78) extends the size laws to all M at once, since nothing in it depends on where a cell sits in the chain.

5.C Technical derivations

5.C.1 Lifetime yield from k founders

We derive (5.74). Start from $V_\infty^{(k)} = \int_0^\infty g_k(t) dt$ with g_k from (5.72), and change variables from t to $x = I(t)$: as t runs $0 \rightarrow \infty$, I runs $1 \rightarrow b$, and since $dI/dt = -\delta J$ we have $dt = -dx/(\delta J)$. The first part of g_k gives

$$\int_0^\infty \delta k K I^{k-1} dt = \int_b^1 k \frac{K}{J} x^{k-1} dx = \int_b^1 k \left(1 + \frac{2\lambda}{\delta} - \frac{2\lambda}{\delta} x\right) x^{k-1} dx,$$

using $K/J = 1 + 2\lambda(1 - I)/\delta$. Integrating,

$$= \left(1 + \frac{2\lambda}{\delta}\right)(1 - b^k) - \frac{2\lambda k}{\delta} \frac{1 - b^{k+1}}{k+1} = (1 - b^k) + \frac{2\lambda}{\delta} \left[\frac{1}{k+1} - b^k + \frac{k b^{k+1}}{k+1}\right],$$

where the second form collects terms over the common denominator $k+1$. The second part of g_k gives, with $J = -\frac{\lambda}{\delta}(I - a)(I - b)$,

$$\int_0^\infty \delta k(k-1) J^2 I^{k-2} dt = \int_b^1 k(k-1) J x^{k-2} dx = k(k-1) \frac{\lambda}{\delta} \int_1^b (x-a)(x-b) x^{k-2} dx,$$

on reversing the limits. Adding the two parts yields (5.74). The formula also holds at $k=1$, the integral term vanishing, and reduces to $V_\infty = (1-b)[1 + \lambda(1-b)/\delta]$ of (5.12). When $\mu = 0$, $b = 0$ and the integral simplifies to $k(k-1) \frac{\lambda}{\delta} \int_1^0 (x-a)x^{k-1} dx = k(k-1) \frac{\lambda}{\delta} \left[\frac{a-1}{k} - \frac{1}{k+1}\right]$, which combines with the first part to give $k + \lambda/\delta$ — the elementary result of (5.75), recovered analytically.

5.C.2 Coefficient extraction for general initial conditions

The general- μ route for the chained model of [section 5.10](#) runs through the k -founder PGF. From [section 5.3.3](#), in the catastrophe convention,

$$G_k(z, t) = \left[\frac{ab\gamma(t) + v(t)z}{1 - \gamma(t)z} \right]^k, \quad \gamma(t) = \frac{1-w}{b-aw}, \quad v(t) = \frac{bw-a}{b-aw}, \quad (5.89)$$

with $w = e^{\theta t}$; note that γ is the sliding ratio P of (5.28). Writing $\Theta_m = \binom{k}{m} (ab)^{k-m} \gamma(t)^{k-m} v(t)^m$ and $\Omega_n = \binom{k-1+n}{n} \gamma(t)^n$, the binomial expansion of the numerator and the negative-binomial

series of $(1 - \gamma z)^{-k}$ give $G_k(z, t) = (\sum_{m=0}^k \Theta_m z^m) (\sum_{n \geq 0} \Omega_n z^n)$, so that the state-probability coefficients are, with H here denoting the power of z rather than the rupture state,

$$[z^H]G_k = \sum_{i=0}^{\min(H,k)} \Theta_{k-i} \Omega_{i+\max(0,H-k)}. \quad (5.90)$$

The chained burst-size laws then require integrals of the form $\int_0^\infty \gamma(t)^n v(t)^m dt$ for arbitrary $n, m \in \mathbb{N}$. Binomial-expanding the two numerators,

$$(1 - w)^n (bw - a)^m = \sum_{i=0}^{n+m} C_i w^i \quad (5.91)$$

for explicit constants C_i , and the problem reduces to the elementary building blocks

$$\int \frac{w^i}{(b - aw)^j} dt = -\frac{i w^i (b - aw)^{1-j}}{b \theta} {}_2F_1\left(1, (1+i) - j; 1+i; \frac{aw}{b}\right), \quad (5.92)$$

whose integral identity is established in [section 1.C](#), at [proposition 1.11](#) and [\(1.127\)](#). The burst-size law for the k th rupture then follows by summing over the initial-condition distribution. For $\mu = 0$ this entire machinery collapses to the geometric decomposition of [section 5.10.1](#); for $\mu > 0$ it remains the available route, and the joint laws are the open problem of [section 5.12.4](#). One step here is not routine: the interchange of summation and integration in [\(5.91\)](#) is justified termwise for t bounded, but in the limit it is asserted rather than proved.

5.C.3 Inter-rupture intervals and the second rupture time

Given k founders, the interval $T(k)$ until rupture decomposes over the holding times: if $G = g$ births precede the catastrophe, the load runs through $k, k+1, \dots, k+g$, so

$$T(k) = \sum_{j=k}^{k+G} \xi_j, \quad \xi_j \sim \text{Exp}((\lambda + \delta)j) \text{ independent}. \quad (5.93)$$

Averaging over $G \sim \text{Geom}_0(s)$, the Laplace transform is

$$\tilde{T}(k, u) = \mathbb{E}[e^{-uT(k)}] = \sum_{g \geq 0} s \rho^g \prod_{j=k}^{k+g} \frac{(\lambda + \delta)j}{(\lambda + \delta)j + u}. \quad (5.94)$$

Writing $u' = u/(\lambda + \delta)$ and unrolling the product with Gamma functions, the series sums to a hypergeometric form — a theme that recurs in the chapter on population dynamics, whose effective parameters are likewise expressed through hypergeometric Laplace transforms:

$$\tilde{T}(k, u) = \frac{s k}{k + u'} {}_2F_1\left(k + 1, 1; k + 1 + u'; \rho\right). \quad (5.95)$$

Differentiating [\(5.94\)](#) at $u = 0$ gives the exact mean

$$\mathbb{E}[T(k)] = \sum_{m \geq 0} \frac{\rho^m}{(\lambda + \delta)(k + m)}, \quad (5.96)$$

which is strictly decreasing in k : the more a cell inherits, the faster it bursts. For $(\lambda, \delta) = (1, 1)$ the first four values are $\mathbb{E}[T(1..4)] \approx 0.693, 0.386, 0.273, 0.212$ at fixed founding loads; the chain intervals T_k , whose founding loads are themselves random, decrease faster still, 0.691, 0.500, 0.377, 0.292 in simulation.

The chain's dependence structure is visible in the second rupture time. It is $t_2 = t_1 + T_2$, but t_1 and T_2 are *not* independent: both are functions of G_1 , the number of births in the first cell. Conditional on $G_1 = g$, however, the two pieces involve disjoint sets of exponential holding times, so their Laplace transforms factor:

$$\mathbb{E}\left[e^{-ut_2} \mid G_1 = g\right] = \Pi(1 + g, u) \tilde{T}(1 + g, u), \quad \Pi(m, u) = \prod_{j=1}^m \frac{(\lambda + \delta)j}{(\lambda + \delta)j + u}. \quad (5.97)$$

Averaging over $G_1 \sim \text{Geom}_0(s)$,

$$\tilde{t}_2(u) = \sum_{g \geq 0} s \rho^g \Pi(1 + g, u) \tilde{T}(1 + g, u). \quad (5.98)$$

The density itself is computable exactly as a mixture of hypoexponential densities, since $t_1 \mid G_1$ and $T_2 \mid (G_1, G_2)$ are sums of independent exponentials, and it is verified against simulation in [section 5.B](#). Moments assemble from the pieces, for instance $\mathbb{E}[t_2] = \mathbb{E}[T(1)] + \sum_{n \geq 1} \mathbb{P}\{r_1 = n\} \mathbb{E}[T(n)]$.

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